Sensors and Actuators Interfacing with Arduino and ESP32

Dr. Hiran Ekanayake

University of Colombo School of Computing

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Sensors and Actuators Interfacing with Arduino and ESP32

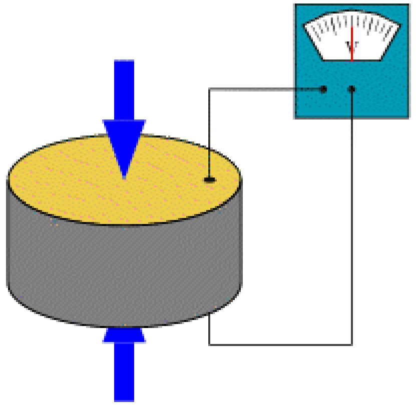
- Arduino (especially the Arduino Uno R3) and ESP32 are among the most popular microcontroller platforms for interfacing with sensors and actuators. **Sensors** allow the microcontroller to perceive the environment, while **actuators** enable it to perform actions based on sensor inputs.

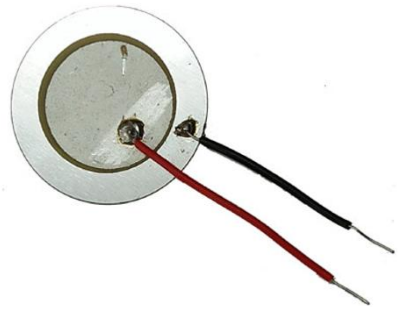
```
Environment
     |
     ▼
+-------------+
|   Sensors   |
+-------------+
     |
     ▼
Signal Conditioning
     |
     ▼
Arduino / ESP32
     |
Decision Making
     |
     ▼
Driver Circuit
     |
     ▼
+--------------+
|  Actuators   |
+--------------+
     |
     ▼
Physical Action
```

Classification of Sensors and Actuators

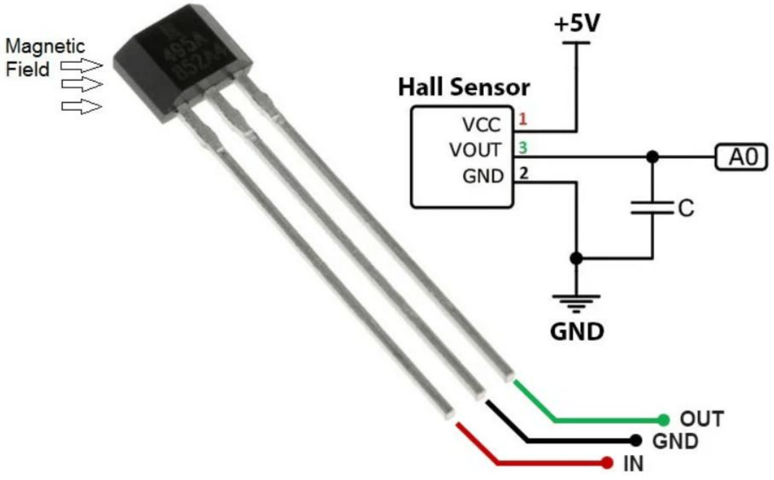
- Sensors and actuators can be classified in several ways depending on their operating principle, output type, application, power source, and method of interaction with the environment. Understanding these classifications helps in selecting suitable devices for Arduino, ESP32, robotics, industrial automation, IoT, and embedded systems.

Sensor Classification



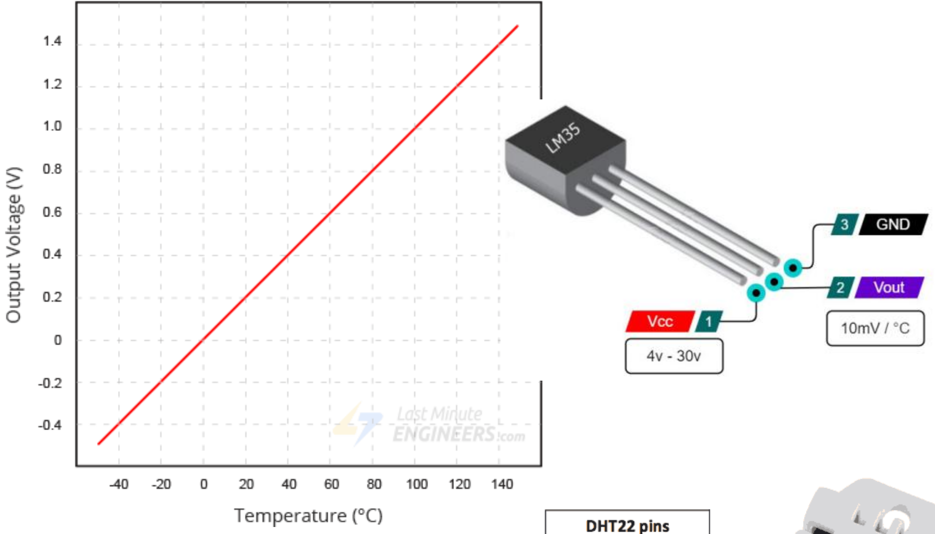


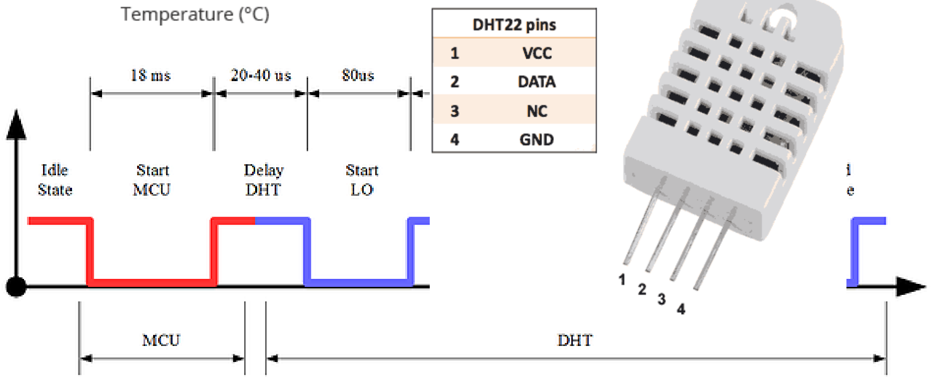
- A. Based on Energy Source
  - **Active Sensors** (Self-generating)
    - Generate electrical signals without requiring external excitation.
    - Examples: Thermocouple, Piezoelectric sensor, Photovoltaic (solar) sensor, Tachogenerator
    - Advantages: No external excitation required, Suitable for remote applications
  - **Passive Sensors** (Require External Power)
    - Need an external voltage or current source.
    - Examples: RTD, Thermistor, LDR, Strain gauge, Hall effect sensor, Capacitive humidity sensor
    - Most Arduino and ESP32 sensors belong to this category.

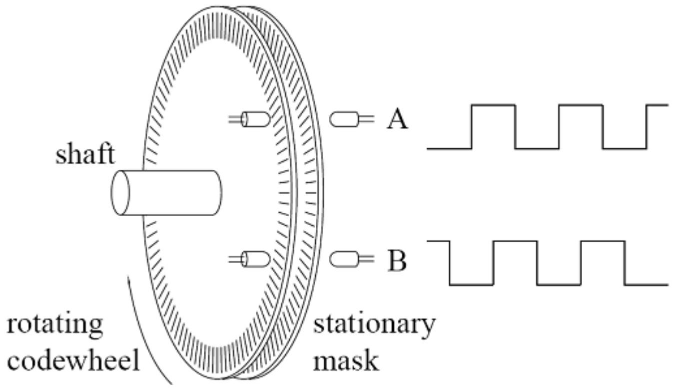


Sensor Classification Cont.

- B. Based on Output Signal
  - **Analog Sensors**
    - Produce continuous signals.
    - Examples: LM35, LDR, Potentiometer, MQ Gas Sensors, Soil moisture sensor, Flex sensor, Pressure transducer
  - **Digital Sensors**
    - Produce binary outputs or communicate digitally.
    - Examples: DHT22, DS18B20, BMP280, PIR, Reed switch, Hall switch
  - **Frequency Output Sensors**
    - Output frequency proportional to measured quantity.
    - Examples: Flow sensors, Rotary encoders, Frequency-output pressure sensors
  - **Pulse Output Sensors**
    - Produce pulses.
    - Examples: Water flow sensor, Rain gauge, Geiger counter

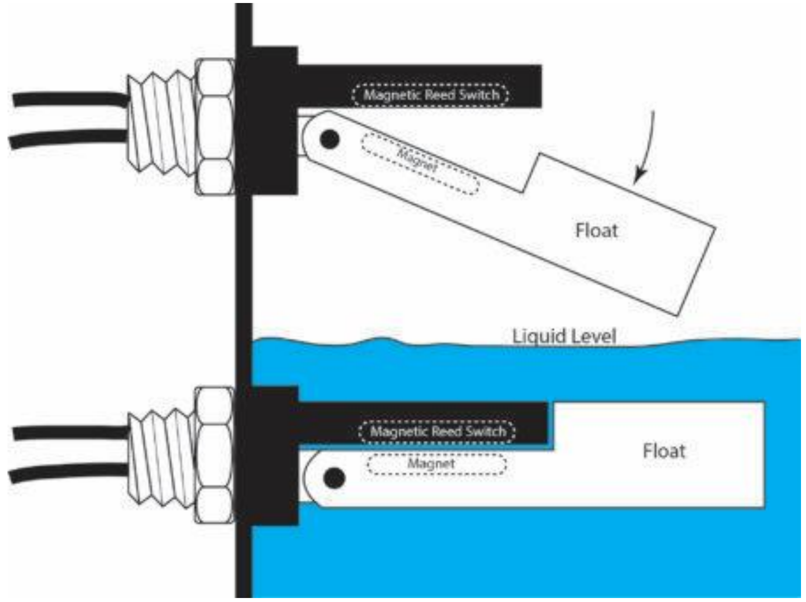


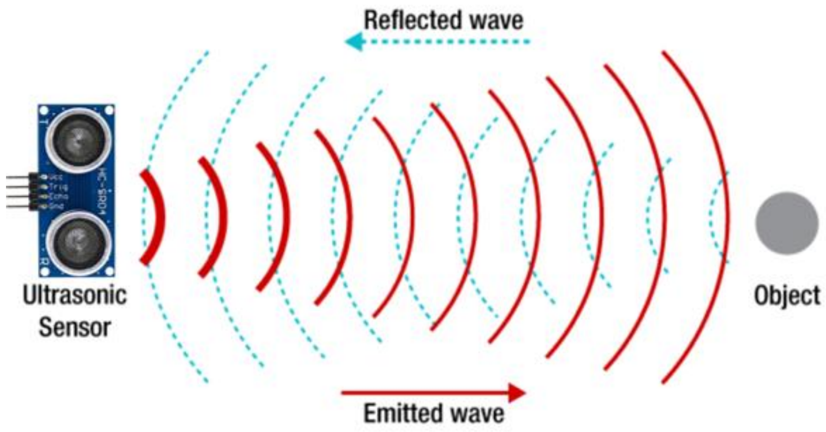




Sensor Classification Cont.

- C. Based on Contact
  - **Contact Sensors**
    - Need physical contact.
    - Examples: Push button, Float switch, Microswitch, Thermistor, Strain gauge
  - **Non-contact Sensors**
    - Detect objects remotely.
    - Examples: Ultrasonic, Infrared, Radar, LiDAR, Camera, Thermal camera





Sensor Classification Cont.

- D. Based on Physical Quantity Measured
  - Temperature Sensors
  - Humidity Sensors
  - Pressure Sensors
  - Force Sensors
  - Position Sensors
  - Distance Sensors
  - Light Sensors
  - Motion Sensors
  - Magnetic Sensors
- Cont.
  - Sound Sensors
  - Gas Sensors
  - Chemical Sensors
  - Electrical Sensors
  - Flow Sensors
  - Level Sensors
  - Image Sensors
  - Biometric Sensors
  - Environmental Sensors

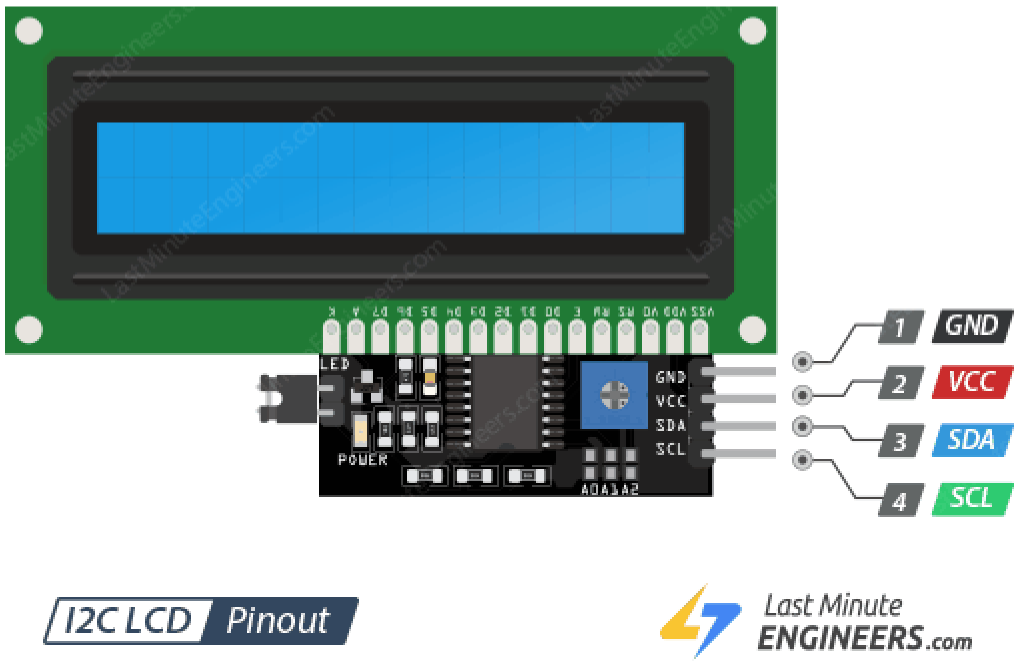
Sensor Classification Cont.

- E. Based on Transduction Principle

Principle	Examples
Resistive	LDR, Thermistor, Strain Gauge
Capacitive	Humidity, Touch sensor
Inductive	Metal detector, Proximity sensor
Piezoelectric	Vibration sensor
Optical	Photodiode, Camera
Magnetic	Hall sensor
Ultrasonic	HC-SR04
Thermal	Thermocouple
Electrochemical	Gas sensor, pH sensor
MEMS	Accelerometer, Gyroscope

Sensor Classification Cont.

- F. Based on Communication Interface
  - Analog
  - GPIO Digital
  - I^2C
  - SPI
  - UART
  - CAN
  - RS-485
  - 1-Wire
  - I^2S



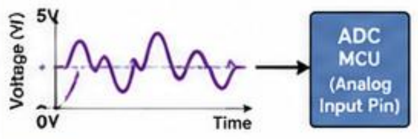
COMMUNICATION INTERFACES

Different Speeds | Multiple Devices | Reliable Communication | Easy Integration | Noise Resistant | Low Power Options

Connect Sensors, Actuators & Modules with Arduino / ESP32 and Other MCUs

1. ANALOG (ADC)

Continuous Voltage Signal

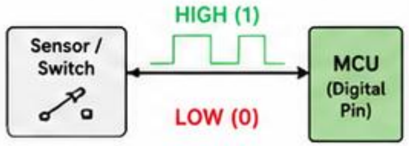


DESCRIPTION
Analog sensors output a continuous voltage that varies with the measured quantity. The MCU converts this voltage to a digital value using an ADC.

WIRING	Signal → Analog Pin
SPEED	–
DISTANCE	Short
EXAMPLES	LM35, LDR, Potentiometer, Microphone
APPLICATIONS	Measuring temperature, light, sound, pressure, humidity, position, etc.

2. GPIO DIGITAL

High / Low Signal

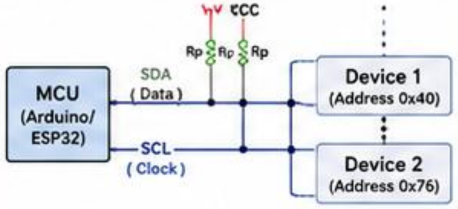


DESCRIPTION
Digital communication uses two states: HIGH (1) or LOW (0). Simple and widely used for switches, buttons, PIR sensors, relays, etc.

WIRING	Signal → Digital Pin
SPEED	Up to ~80 MHz (MCU dependent)
DISTANCE	Short
EXAMPLES	Push Button, PIR, Relay, Hall Sensor, Limit Switch, IR Sensor
APPLICATIONS	On/Off control, status signals, event detection

3. I²C (Inter-Integrated Circuit)

Two-Wire, Multi-Master



DESCRIPTION
I²C uses only two lines (SDA, SCL). Multiple devices share the same bus using unique addresses.

WIRING	SDA, SCL, GND, VCC (with pull-up resistors)
SPEED	100 kbps (Std), 400 kbps (Fast), 1 Mbps (Fm+)
DISTANCE	Up to ~1 m (short), can extend with buffers
EXAMPLES	OLED, LCD, BMP280, BME280, RTC, EEPROM, MPU6050
APPLICATIONS	Sensor networks, displays, data logging

4. SPI (Serial Peripheral Interface)

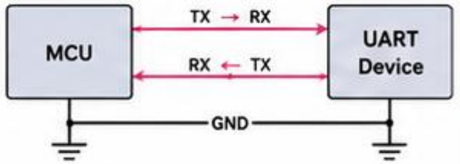
High-Speed, Full-Duplex

DESCRIPTION
SPI uses four main lines and a Chip Select (CS) line per device. High speed and full duplex communication.

WIRING	SCK, MOSI, MISO, CS, GND, VCC
SPEED	Up to tens of MHz (MCU dependent)
DISTANCE	Short (PCB traces / short cables)
EXAMPLES	SD Card, TFT Display, RFID (MFRC522), MAX7219, ADC (MCP3008)
APPLICATIONS	High-speed data transfer, displays, memory devices

5. UART (Universal Asynchronous Receiver Transmitter)

Asynchronous Serial Communication

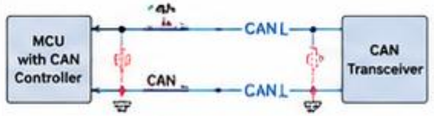


DESCRIPTION
UART uses two signal lines (TX, RX) and GND. Asynchronous communication – no clock line.

WIRING	TX, RX, GND
SPEED	Typical 9600 bps – 921600 bps+
DISTANCE	Short (TTL), Long with converters
EXAMPLES	GPS, GSM, Bluetooth (HC-05), LoRa Modules, Serial GPS
APPLICATIONS	Wireless modules, GPS, debugging, serial communication

6. CAN (Controller Area Network)

Robust, Multi-Master Bus

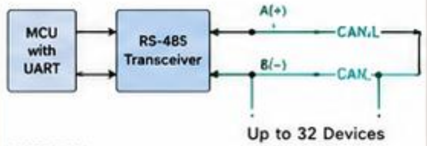


DESCRIPTION
CAN is a differential two-wire bus designed for robust communication in noisy environments.

WIRING	CAN_H, CAN_L, GND (120Ω termination)
SPEED	Up to 1 Mbps (bus length dependent)
DISTANCE	Up to 40 m @ 1 Mbps, up to 1 km @ 50 kbps
EXAMPLES	CAN Bus Modules (MCP2515 + TJA1050), Automotive ECUs
APPLICATIONS	Automotive, Industrial control, Robotics, Real-time systems

7. RS-485

Differential, Long Distance

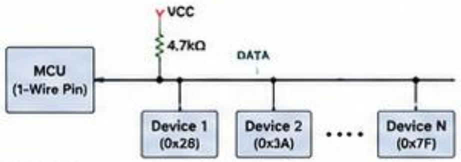


DESCRIPTION
RS-485 is a differential multi-drop network widely used in industrial applications.

WIRING	A(+), B(–), GND
SPEED	Up to 10 Mbps
DISTANCE	Up to 1200 m
EXAMPLES	MAX485, SN75176 Modules
APPLICATIONS	Industrial automation, Modbus RTU, sensor networks

8. 1-Wire

Single Data Line



DESCRIPTION
1-Wire uses a single data line and GND. Devices have unique 64-bit addresses.

WIRING	DATA, GND (Pull-up resistor to VCC)
SPEED	16.3 kbps (standard)
DISTANCE	Up to ~30 m
EXAMPLES	DS18B20, EEPROM (DS2431), 1-Wire Key
APPLICATIONS	Temperature logging, identification, small sensor networks

9. I²S (Inter-IC Sound)

Digital Audio Interface

DESCRIPTION
I²S is a serial bus interface for digital audio data transfer between ICs.

WIRING	SCK, WS, SD (Out), SD (In), MCLK, GND
SPEED	Up to tens of MHz (audio sample rate)
DISTANCE	Short (on-board / short cables)
EXAMPLES	MEMS Microphones (INMP441), Audio DAC (PCM5102), MAX98357A
APPLICATIONS	Digital microphones, audio playback, voice capture, audio processing

INTERFACE COMPARISON

ANALOG	Continuous signal (0–5V / 0–3.3V)
GPIO DIGITAL	HIGH / LOW (0 or 1)
I²C	2 wires, Multi-device Addressed
SPI	High speed, Full duplex 4+ wires
UART	2 wires, Asynchronous Point-to-point
CAN	Differential, Robust Multi-master
RS-485	Differential, Long distance Multi-drop
1-WIRE	Single data line Addressable devices
I²S	Digital audio High quality

TIPS

- Choose the right interface based on speed, distance, number of devices and application.
- Use pull-up resistors for I²C and 1-Wire.

GOOD PRACTICES

- Keep wires short for high speed buses (SPI, I²S).
- Use termination resistors for CAN and RS-485.

COMMON USE CASES

Sensors & Actuators | Displays | Data Logging | Industrial Control | Robotics | Audio Devices | IoT Communication

Different interfaces, endless possibilities!

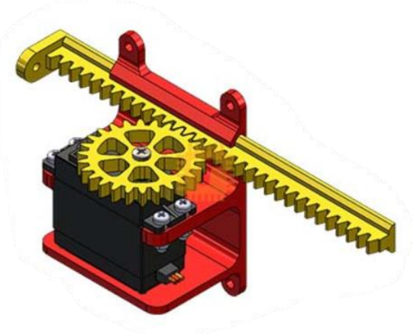
Sensor Classification Cont.

- G. Based on Application
  - Industrial sensors
  - Automotive sensors
  - Medical sensors
  - Agricultural sensors
  - Consumer electronics sensors
  - Aerospace sensors
  - Smart home sensors
  - Robotics sensors
  - IoT sensors
  - Wearable sensors

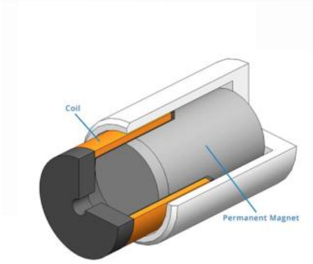
Actuator Classification

- Actuators convert electrical, pneumatic, hydraulic, or thermal energy into physical action.
- A. Based on Energy Source
  - **Electrical Actuators**
    - Examples: DC motor, Servo motor, Stepper motor, Solenoid, Relay, LED, Speaker
  - **Pneumatic Actuators**
    - Operate using compressed air. Examples: Pneumatic cylinder, Air gripper, Air motor
  - **Hydraulic Actuators**
    - Operate using pressurized fluid. Used in heavy machinery.
    - Examples: Hydraulic cylinder, Hydraulic motor
  - **Thermal Actuators**
    - Examples: Heater, Peltier module, Shape Memory Alloy (SMA)
  - **Piezoelectric Actuators**
    - Examples: Precision positioning stages, Ultrasonic transducers, Inkjet printer heads

Actuator Classification Cont.

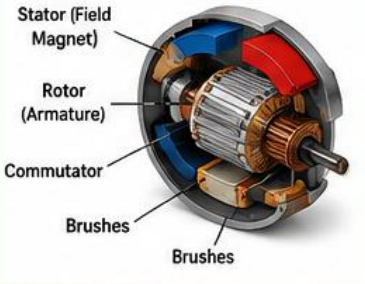


- B. Based on Motion
  - **Linear Actuators**
    - Produce straight-line motion.
    - Examples: Solenoid, Hydraulic cylinder, Pneumatic cylinder, Electric linear actuator
  - **Rotary Actuators**
    - Produce rotational motion.
    - Examples: DC motor, Servo motor, Stepper motor, AC motor, Brushless DC motor (BLDC)
  - **Oscillatory Actuators**
    - Move back and forth.
    - Examples: Voice coil actuator, Vibrating motor

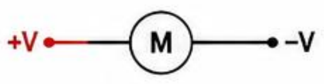


TYPES OF MOTORS – VISUAL DIAGRAM

1. DC MOTOR (Brushed)



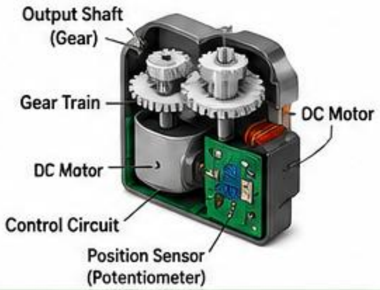
WIRING DIAGRAM



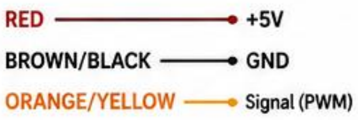
- Operates on DC supply
- Uses brushes and commutator
- Simple and inexpensive
- Speed control: Easy

Typical Uses:
Toys, Fans, Small robots, Power windows, DIY projects

2. SERVO MOTOR



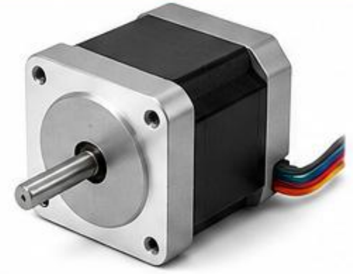
WIRING DIAGRAM (3 WIRES)

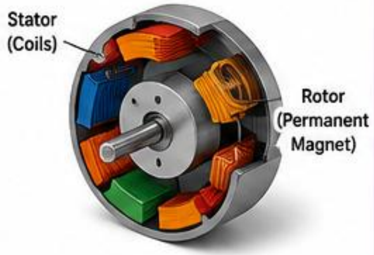


- Built-in control circuit
- Precise position control
- Controlled by PWM signal
- Limited rotation (~180°)

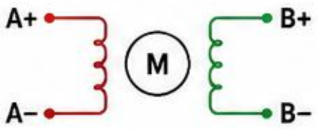
Typical Uses:
Robotics, RC cars, Drones, Camera gimbals, Automation

3. STEPPER MOTOR





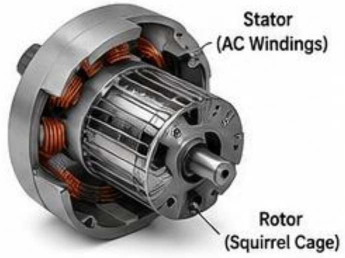
WIRING DIAGRAM (Example: 4 Wires)



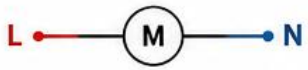
- Moves in precise steps
- Excellent for positioning
- No feedback required
- Higher torque at low speed

Typical Uses:
3D Printers, CNC machines, Plotters, Robotics, Automation

4. AC MOTOR (Induction Motor)



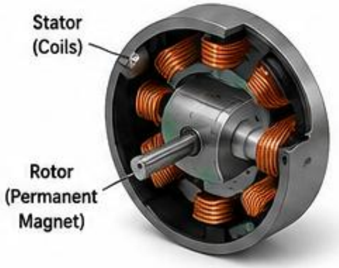
WIRING DIAGRAM (Single Phase)



- Operates on AC supply
- Robust and reliable
- High power capability
- Speed control: VFD required

Typical Uses:
Industrial machines, Pumps, Fans, Compressors, Conveyors

5. BRUSHLESS DC MOTOR (BLDC Motor)



WIRING DIAGRAM (3 Wires to ESC)

- No brushes, high efficiency
- High speed and torque
- Long life, low maintenance
- Controlled by ESC (electronic)

Typical Uses:
Drones, Electric vehicles, RC planes, High performance apps

Actuator Classification Cont.

- C. Based on Control Method
  - **On-Off Actuators**
    - Examples: Relay, Solenoid, LED, Buzzer
  - **Proportional Actuators**
    - Output varies continuously.
    - Examples: Servo motor, BLDC motor, Proportional valve, Dimmable LED

Actuator Classification Cont.

- D. Based on Operating Principle

Principle	Examples
Electromagnetic	Relay, Solenoid
Electromechanical	Motor
Hydraulic	Hydraulic cylinder
Pneumatic	Pneumatic cylinder
Thermal	Heater, Peltier
Piezoelectric	Piezo actuator
Electrostatic	MEMS actuators
Shape Memory Alloy	Nitinol actuator

Actuator Classification Cont.

- E. Based on Function
  - Visual Actuators
    - LED, RGB LED, LCD, OLED, Seven-segment display
  - Audio Actuators
    - Buzzer, Speaker, Siren
  - Motion Actuators
    - Servo, Stepper, DC motor, BLDC motor, Linear actuator
  - Fluid Control Actuators
    - Solenoid valve, Motorized valve, Pump
  - Switching Actuators
    - Relay, Solid State Relay (SSR), Contactor
  - Heating/Cooling Actuators
    - Heating element, Peltier cooler, Fan

Actuator Classification Cont.

- F. Based on Communication
  - GPIO-controlled
  - PWM-controlled
  - Analog-controlled
  - I^2C-controlled (e.g., PWM driver boards)
  - SPI-controlled
  - UART-controlled
  - CAN-controlled
  - Ethernet/Industrial fieldbus-controlled

Common Sensor & Actuator Modules

Note: The diagrams shown are intended for illustrative purposes only and may not accurately represent the actual appearance of the sensors and actuators. Likewise, the specifications, descriptions, and prices provided are approximate and may not be 100% accurate. Always refer to the manufacturer's datasheet and official documentation for precise technical information.

TEMPERATURE SENSORS

Popular Temperature Sensor Modules for Arduino / ESP32 Projects

Accurate Measurement | Easy to Interface | Reliable & Durable | Wide Range of Applications

	LM35 MODULE	TMP36	DS18B20 WATERPROOF	MAX6675 + K-TYPE THERMOCOUPLE	MAX31855 THERMOCOUPLE AMPLIFIER
					
	Linear temperature sensor with analog output	Low voltage, low power analog temperature sensor	Digital temperature sensor with 1-Wire interface	Thermocouple amplifier for high temperature measurement	High accuracy thermocouple amplifier with advanced features
Sensor / IC	LM35	TMP36	DS18B20	MAX6675	MAX31855
Interface	Analog	Analog	1-Wire	SPI	SPI
Voltage Supply	2.7V – 5.5V DC	2.7V – 5.5V DC	3.0V – 5.5V DC	3.0V – 5.5V DC	3.0V – 5.5V DC
Temperature Range	-55°C to +150°C	-40°C to +125°C	-55°C to +125°C	0°C to +1024°C	-270°C to +1372°C
Accuracy	±0.5°C @ 25°C	±0.5°C @ 25°C	±0.5°C (-10°C to +85°C)	±2°C (0°C to +700°C)	±0.25% (typ.)
Resolution	10 mV / °C	10 mV / °C	9 to 12-bit (0.0625°C)	0.25°C	0.25°C
Features	• Linear output (10 mV/°C) • No external calibration • Low self-heating • Easy to use	• Low power consumption • Good linearity • Stable and reliable	• Waterproof probe • Unique 64-bit address • Multi-drop (many sensors on single pin) • No external components	• Works with K-Type thermocouple • High temperature range • Cold-junction compensation • SPI digital output	• Very high accuracy • Wide temperature range • Built-in fault detection • Cold-junction compensation • Open thermocouple detection
Approx. Price	$1 – $3 (Rs. 350 – 1,000)	$2 – $4 (Rs. 700 – 1,400)	$2 – $5 (Rs. 700 – 1,400)	$6 – $12 (Rs. 2,100 – 4,300)	$10 – $18 (Rs. 3,500 – 6,500)

COMPARISON AT A GLANCE

	Temp. Range	Accuracy (typ.)	Interface
LM35 Module	-55°C to +150°C	±0.5°C	Analog
TMP36	-40°C to +125°C	±0.5°C	Analog
DS18B20 Waterproof	-55°C to +125°C	±0.5°C	1-Wire
MAX6675 + K-Type	0°C to +1024°C	±2°C	SPI
MAX31855	-270°C to +1372°C	±0.25%	SPI

APPLICATIONS

Weather Stations | Industrial Monitoring | HVAC Systems | Cooking & Ovens | 3D Printers & Makers

TIPS:
- For general purpose and low cost projects: LM35 or TMP36
- For long distance / waterproof applications: DS18B20
- For high temperature applications: MAX6675 or MAX31855

Note: Always follow wiring diagram and use proper voltage supply. Use filters and shielding for high noise environments.

HUMIDITY SENSORS

Popular Digital Humidity Sensors for Arduino / ESP32 / IoT Projects

- Measure Relative Humidity (RH)
- High Accuracy & Stability
- Low Power Consumption
- Easy Interface (Digital/I²C)

DHT11

Low-cost Digital Humidity & Temperature Sensor

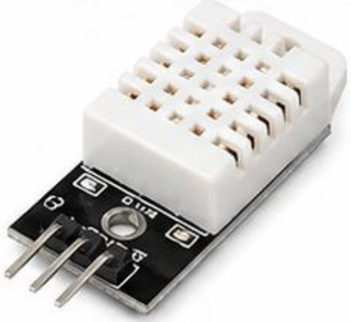
FEATURES

- Humidity Range: 20 – 90 %RH
- Accuracy: ±5 %RH
- Temperature Range: 0 – 50 °C
- Resolution: 1 %RH, 1 °C
- Interface: Single-Wire Digital
- Operating Voltage: 3.3 – 5.5 V
- Low cost, simple to use

Approx. Price: **$2 – $3**

DHT22 (AM2302)

High Accuracy Digital Humidity & Temperature Sensor



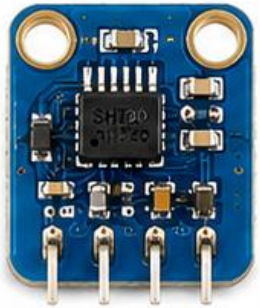
FEATURES

- Humidity Range: 0 – 100 %RH
- Accuracy: ±2 %RH (typ.)
- Temperature Range: -40 – 80 °C
- Resolution: 0.1 %RH, 0.1 °C
- Interface: Single-Wire Digital
- Operating Voltage: 3.3 – 6 V
- More accurate & wider range

Approx. Price: **$4 – $8**

SHT20

Digital Humidity & Temperature Sensor (I²C)



FEATURES

- Humidity Range: 0 – 100 %RH
- Accuracy: ±3 %RH
- Temperature Range: -40 – 125 °C
- Resolution: 0.04 %RH, 0.01 °C
- Interface: I²C (Addr: 0x40)
- Operating Voltage: 2.4 – 5.5 V
- Fast response, low power

Approx. Price: **$5 – $8**

SHT31

High Accuracy & Fast Response Humidity Sensor (I²C)

FEATURES

- Humidity Range: 0 – 100 %RH
- Accuracy: ±2 %RH (typ.)
- Temperature Range: -40 – 125 °C
- Resolution: 0.01 %RH, 0.01 °C
- Interface: I²C (Addr: 0x44)
- Operating Voltage: 2.4 – 5.5 V
- Excellent stability & reliability

Approx. Price: **$6 – $12**

AHT20

Low Power Digital Humidity & Temperature Sensor (I²C)

FEATURES

- Humidity Range: 0 – 100 %RH
- Accuracy: ±3 %RH
- Temperature Range: -40 – 85 °C
- Resolution: 0.024 %RH, 0.01 °C
- Interface: I²C (Addr: 0x38)
- Operating Voltage: 2.2 – 5.5 V
- Very low power consumption

Approx. Price: **$3 – $6**

Applications

Weather stations, Home automation, Greenhouses, IoT devices, Data loggers, HVAC systems, Smart agriculture and more.

Comparison at a Glance

- DHT11 – Best for beginners & low-cost projects
- DHT22 – Good accuracy and wide range
- SHT20 – Faster response with I²C interface
- SHT31 – Highest accuracy & stability
- AHT20 – Low power & compact size

Interface Guide

DHT11 / DHT22 → Single-Wire Digital

SHT20 / SHT31 / AHT20 → I²C (SCL, SDA)

Scan for Datasheets & More Info

TEMPERATURE + HUMIDITY + PRESSURE SENSORS

Compact, Accurate and Reliable Environmental Sensing for Arduino, ESP32 & IoT Projects

BME280
Environmental Sensor

- Temperature
- Humidity
- Pressure

FEATURES

- Measures Temperature, Humidity & Pressure
- High accuracy and long term stability
- Low power consumption (3.3 µA)
- Wide supply voltage: 1.8V – 3.6V
- Small size: 2.5 x 2.5 x 0.93 mm

T Accuracy	RH Accuracy	Pressure Accuracy
±1.0 °C	±3%	±1 hPa

Interface	Approx. Price
I²C / SPI	USD 5 – 10

BME680
Environmental Sensor

- Temperature
- Humidity
- Pressure
- Gas / VOC

FEATURES

- Temperature, Humidity, Pressure + Gas/VOC
- Integrated gas sensor for air quality (IAQ)
- Detects VOC, CO, ethanol, NH_3, etc.
- Low power with advanced algorithms
- Ideal for indoor air quality monitoring

T Accuracy	RH Accuracy	Pressure Accuracy	Gas
±1.0 °C	±3%	±1 hPa	Up to 500 °C

Interface	Approx. Price
I²C / SPI	USD 10 – 18

BMP280
Pressure + Temperature Sensor

- Temperature
- Pressure

FEATURES

- Measures Pressure and Temperature
- High precision and excellent linearity
- Low power consumption (2.7 µA)
- Wide pressure range: 300 – 1100 hPa
- Ideal for altitude and weather apps

T Accuracy	Pressure Accuracy
±1.0 °C	±1 hPa

Interface	Approx. Price
I²C / SPI	USD 3 – 6

BMP180
Pressure + Temperature Sensor

- Temperature
- Pressure

FEATURES

- Measures Pressure and Temperature
- Very low power and easy to use
- Pressure range: 300 – 1100 hPa
- Good accuracy for general applications
- Great for weather stations & drones

T Accuracy	Pressure Accuracy
±1.0 °C	±1 hPa

Interface	Approx. Price
I²C	USD 2 – 4

APPLICATIONS

- Weather Stations
- Drones & UAVs
- IoT Monitoring
- Smart Home Automation
- Agriculture Monitoring
- Data Loggers & Research

Why Choose These Sensors?

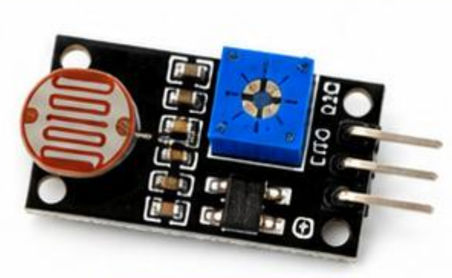
- Compact size and easy integration
- Low power for battery operated systems
- High accuracy and long term stability
- Works with Arduino, ESP32, Raspberry Pi and more

LIGHT SENSORS

Measure Light Intensity, Color & Ambient Illumination

Detect Light | High Accuracy | Low Power Consumption | Easy to Interface | Wide Applications | Energy Efficient

LDR (Photoresistor)



ANALOG (Resistive)

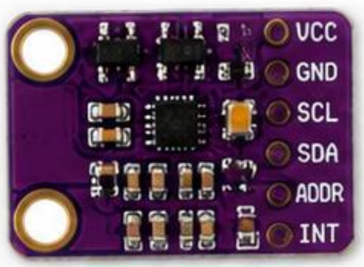
FEATURES
- Simple and Low Cost
- Wider detection range
- High sensitivity to visible light
- Can be used with voltage divider
- Analog output (resistance changes with light)

Light Range : ~1 – 100,000 Lux
Operating Voltage : 3.3V – 5V

APPLICATIONS
- Street light control
- Automatic brightness control
- Day/Night detection

Approx. Price: $0.5 – $2

BH1750



DIGITAL (I²C)

FEATURES
- Digital ambient light sensor
- High accuracy and resolution (16-bit)
- Wide measurement range
- Built-in ADC
- I²C interface (3.3V / 5V compatible)

Light Range : 1 – 65,535 Lux
Operating Voltage : 2.4V – 3.6V

APPLICATIONS
- Display backlight control
- Camera exposure control
- Indoor light measurement

Approx. Price: $2 – $4

TSL2561

DIGITAL (I²C)

FEATURES
- High precision light-to-digital converter
- Wide dynamic range (up to 100,000:1)
- Measures full spectrum IR & visible light
- Programmable gain and timing
- Interrupt output available

Light Range : 0.1 – 40,000 Lux
Operating Voltage : 2.7V – 3.6V

APPLICATIONS
- Smart home lighting
- Industrial light monitoring
- Lux meter, data loggers

Approx. Price: $5 – $8

TCS3200

DIGITAL (Frequency Output)

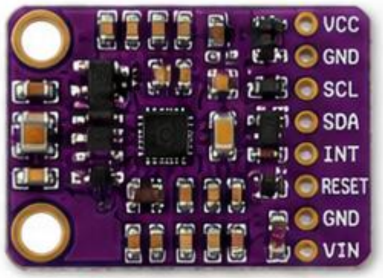
FEATURES
- Color light sensor (RGB)
- Detects Red, Green, Blue components
- Frequency output proportional to light intensity
- Adjustable sensitivity
- Works with or without IR filter

Light Range : Depends on Color & Fliter
Operating Voltage : 2.7V – 5V

APPLICATIONS
- Color detection and sorting
- RGB LED control
- Robotics, education projects

Approx. Price: $3 – $6

APDS9960



DIGITAL (I²C)

FEATURES
- 4-in-1 Sensor (Ambient Light, RGB, Proximity, Gesture)
- Built-in IR LEDs for proximity & gesture
- High sensitivity & low power
- I²C interface
- Programmable interrupts

Light Range : 0 – 65,535 Lux
Operating Voltage : 2.4V – 3.6V

APPLICATIONS
- Gesture recognition
- Touchless controls
- Mobile & IoT devices

Approx. Price: $6 – $10

DID YOU KNOW?
Light sensors are used in automatic street lights, smartphone brightness control, camera exposure, solar trackers, smart home systems and many more applications.

INTERFACES EXPLAINED
Analog: Outputs a variable voltage | **I²C:** Two-wire digital communication (SDA, SCL)
Frequency Output: Outputs a square wave whose frequency changes with light intensity

TIPS
Use proper libraries, decoupling capacitors and correct voltage levels for accurate readings.

DISTANCE / PROXIMITY SENSORS

Types, Features & Applications

Distance and proximity sensors detect the presence or distance of objects without (or with minimal) physical contact.

1. HC-SR04 (Ultrasonic Sensor)

Range: 2 – 400 cm

Interface: Digital (Trig / Echo)

FEATURES

- Low cost
- Easy to use
- 15° measuring angle
- Works in most lighting conditions

APPLICATIONS

- Obstacle detection
- Robot navigation
- Distance measurement
- Parking assist

Price: USD $2 – $4

2. JSN-SR04T (Waterproof Ultrasonic)

Range: 20 – 600 cm

Interface: Digital (Trig / Echo)

FEATURES

- Waterprooff (IP68)
- Longer range
- High accuracy
- Suitable for outdoor use

Applications

- Tank level measurement
- Outdoor robot navigation
- Flood monitoring
- Industrial automation

Price: USD $5 – $8

3. VL53L0X (Time-of-Flight ToF)

Range: 2 cm – 2 m

Interface: I°C

FEATURES

- Time-of-Flight technology
- High accuracy (±3%)
- Small size
- Immune to IR interference

Applications

- Gesture detection
- Robot obstacle avoidance
- Presence detection
- Level sensing

Price: USD $5 – $10

4. VL53L1X (Advanced ToF Sensor)

Range: 4 cm – 4 m

Interface: I°C

FEATURES

- Longer range than VL53L0X
- High speed (up to 400 kHz)
- Multi-zone detection
- Better ambient light immunity

Applications

- Robotics
- People counting
- Level measurement
- Industrial automation

Price: USD $10 – $20

5. Sharp GP2Y0A21YK0F (IR Distance Sensor)

Range: 10 – 80 cm

Interface: Analog

FEATURES

- IR triangulation method
- Compact size
- Good linearity
- Low power consumption

APPLICATIONS

- Robot navigation
- Industrial equipment
- Object detection
- Level sensing

Price: USD $10 – $15

6. TFmini LiDAR Module (Laser Distance Sensor)

Range: 0.1 – 12 m

Interface: UART (Serial)

FEATURES

- Laser (ToF) measurement
- High accuracy (±2 cm)
- Long range up to 12 m
- Compact and lightweight

APPLICATIONS

- Robot mapping
- Obstacle detection
- Drones
- Industrial automation

Price: USD $35 – $50

QUICK COMPARISON

Technology	Ultrasonic	Ultrasonic (Waterproof)	Time-of-Flight (ToF)	Advanced ToF	Infrared (Analog)	LiDAR (Laser ToF)
Typical Range	2 – 400 cm	20 – 600 cm	2 cm – 2 m	4 cm – 4 m	10 – 80 cm	0.1 – 12 m
Accuracy	~ 3 mm	~ 5 mm	± 3%	± 2 cm	~ 1 cm	± 2 cm
Best For	Low cost projects, general purpose	Outdoor & waterproof applications	Short range, compact & precise measurement	Longer range ToF & multi-zone detection	Analog output & compact designs	Long range, high precision applications

TIPS

- Use ultrasonic sensors for general purpose and cost sensitive projects.
- Use ToF sensors (VL53L0X / VL53L1X) for accurate and compact distance measurement.
- Use LiDAR (TFmini) for long range and high precision applications.
- Avoid using ultrasonic sensors in soft/fabric or very noisy environments.
- Check ambient light conditions when using IR sensors.

MOTION SENSORS

Detect Movement, Motion, Orientation and Vibration for Smart Projects

Why Motion Sensors?
- Detect presence, movement & motion
- Improve automation & energy efficiency
- Essential for Robotics, IoT & Security
- Easy to interface with Arduino / ESP32

1. PIR MOTION SENSOR (HC-SR501)

Detects movement of people or animals by sensing infrared radiation changes.

FEATURES
- **Detection Range:** 3 – 7 meters
- **Detection Angle:** ~120°
- **Operating Voltage:** 4.5 – 20V DC
- **Output:** Digital (HIGH/LOW)
- **Typical Current:** 50 – 80 µA
- **Applications:** Security systems, Lighting control, Automation

Approx. Price: $2 – $4

2. MICROWAVE RADAR SENSOR (RCWL-0516)

Uses Doppler radar to detect movement through objects like plastic, glass or thin walls.

FEATURES
- **Detection Range:** 1 – 7 meters
- **Operating Voltage:** 4 – 28V DC
- **Output:** Digital (HIGH/LOW)
- Adjustable Sensitivity and Hold Time
- Works through non-metallic materials
- **Applications:** Touchless switches, Smart lighting, Occupancy detection

Approx. Price: $2 – $5

3. VIBRATION SENSOR (SW-420)

Detects vibrations or shocks from moving or vibrating objects.

FEATURES
- **Detection Type:** Vibration / Shock
- **Operating Voltage:** 3.3 – 5V DC
- **Output:** Digital (HIGH/LOW)
- **Sensitivity:** Adjustable
- Onboard LED Indicator
- **Applications:** Alarm systems, Robot collision, Machine monitoring

Approx. Price: $1 – $3

4. TILT SENSOR (KY-020 / SW-520D)

Detects tilting or orientation change from its reference position.

FEATURES
- **Tilt Angle:** ~10 – 15°
- **Operating Voltage:** 3.3 – 5V DC
- **Output:** Digital (HIGH/LOW)
- Small Size & Easy to use
- Onboard LED Indicator
- **Applications:** Anti-theft devices, Level detection, Robotics

Approx. Price: $1 – $2

5. 6-AXIS IMU (MOTION MODULE) (MPU6050)

Combines 3-axis accelerometer and 3-axis gyroscope to detect motion, rotation and orientation.

FEATURES
- **Accelerometer Range:** ±2g, ±4g, ±8g, ±16g
- **Gyroscope Range:** ±250, ±500, ±1000, ±2000 °/s
- **Interface:** I2C (Default) / SPI
- **Operating Voltage:** 3.3 – 5V DC
- **Applications:** Robotics, Drones, Gesture control, Stabilization

Approx. Price: $3 – $6

COMMON APPLICATIONS

Security Systems | Automatic Lighting | Alarm Systems | Robotics & Automation | Smart Home Projects | Occupancy Detection | Drones & Navigation | Condition Monitoring

INTERFACING
- Digital Output → Connect to GPIO pins
- Analog Output → Read via ADC
- I2C / SPI → Connect to respective pins
- Compatible with Arduino, ESP32, Raspberry Pi

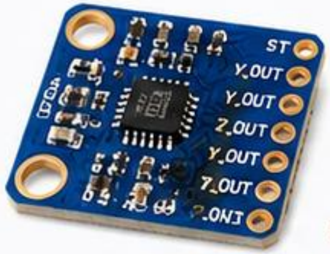
Note: Prices are approximate and may vary by region and supplier.

IMU (ACCELEROMETER / GYROSCOPE) SENSORS

Measure Motion, Orientation & Acceleration in 3D Space

Drones | Robotics | Vehicle Stability | Motion Tracking

ADXL335
3-Axis Analog Accelerometer



Low cost, analog output accelerometer for basic motion and tilt measurements.

Type	Accelerometer
Axes	3-Axis (X, Y, Z)
Range	±3 g
Output	Analog Voltage
Supply Voltage	1.8V – 3.6V
Interface	Analog
Noise	220 µg/√Hz

Features
- Wide supply range (1.8V – 3.6V)
- Low power consumption (~350 µA)
- Analog outputs for X, Y, Z axes
- Small size and easy to use

Applications
- Tilt detection
- Vibration measurement
- Simple motion sensing

$3 – $5

ADXL345
3-Axis Digital Accelerometer

High resolution digital accelerometer with advanced features and low power.

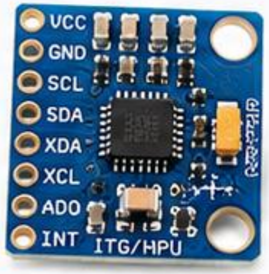
Type	Accelerometer
Axes	3-Axis (X, Y, Z)
Range	±2 / ±4 / ±8 / ±16 g
Output	Digital (16-bit)
Supply Voltage	2.0V – 3.6V
Interface	I²C / SPI
Noise	~220 µg/√Hz

Features
- High resolution (13-bit)
- FIFO buffer, Activity/Inactivity detection
- Tap/Double Tap detection
- Free-fall detection

Applications
- Motion logging
- Step counter / Pedometer
- Vibration analysis

$4 – $7

MPU6050
6-Axis Accelerometer + Gyroscope



Popular 6-axis IMU combining a 3-axis accelerometer and 3-axis gyroscope.

Type	Accelerometer + Gyroscope
Axes	6-Axis (3 Accel + 3 Gyro)
Accel Range	±2 / ±4 / ±8 / ±16 g
Gyro Range	±250 / ±500 / ±1000 / ±2000 °/s
Output	Digital (16-bit)
Supply Voltage	3.0V – 5.0V
Interface	I²C
Features	DMP (Motion Processing Unit) Built-in Temperature Sensor

- 6-axis motion tracking
- DMP for sensor fusion (Quaternion, Euler)
- Low power consumption
- Widely supported by libraries

Applications
- Drones & RC Aircraft
- Robotics & Balancing
- Motion tracking & VR

$3 – $6

MPU9250
9-Axis IMU (Accel + Gyro + Magnetometer)

Advanced 9-axis IMU with AK8963 magnetometer for orientation and heading.

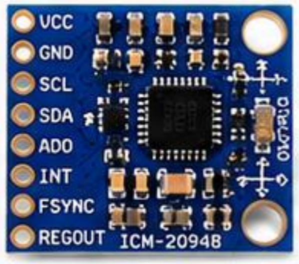
Type	9-Axis IMU
Axes	3 Accel + 3 Gyro + 3 Mag
Accel Range	±2 / ±4 / ±8 / ±16 g
Gyro Range	±250 / ±500 / ±1000 / ±2000 °/s
Mag Range	±4800 µT
Output	Digital (16-bit)
Supply Voltage	3.0V – 5.0V
Interface	I²C / SPI

Features
- 9-axis sensor fusion
- Integrated AK8963 magnetometer
- Excellent for orientation & navigation
- Low power with multiple modes

Applications
- Drones, UAVs & Navigation
- Robotics & SLAM
- AR/VR & Motion Capture

$8 – $15

ICM-20948
9-Axis IMU (High Performance)



High performance 9-axis IMU with low noise and high accuracy.

Type	9-Axis IMU
Axes	3 Accel + 3 Gyro + 3 Mag
Accel Range	±2 / ±4 / ±8 / ±16 g
Gyro Range	±250 / ±500 / ±1000 / ±2000 °/s
Mag Range	±4900 µT
Output	Digital (16-bit)
Supply Voltage	2.5V – 3.3V
Interface	I²C / SPI

Features
- Very low noise & high stability
- Advanced sensor fusion (DMP)
- Supports up to 1125 Hz data rate
- Excellent temperature stability

Applications
- High end drones & gimbals
- Industrial & Automotive
- Precision Motion Tracking

$10 – $20

COMMON IMU TERMS
- Accelerometer : Measures acceleration due to gravity or motion (in g)
- Gyroscope : Measures angular velocity (in °/s)
- Magnetometer : Measures magnetic field (in µT) for heading
- IMU : Combination of sensors for complete motion tracking

INTERFACE TYPES
- I²C — 2-Wire Interface, Easy to use
- SPI — High Speed Interface
- Analog — Analog Output

TYPICAL APPLICATIONS
Drones & UAVs | Robotics | Wearables | Industrial Automation | Balance Systems | Navigation & Mapping

Scan for Datasheets & Projects

MAGNETIC SENSORS

Detect Magnetic Fields | Position | Speed | Direction

1. HALL EFFECT SENSOR (A3144 / 49E / SS49E)

Detects magnetic field using Hall Effect.

FEATURES

- Detects presence of magnetic field
- Digital Output (High / Low)
- Operates from 2.5V – 24V
- Low power consumption
- Fast response
- Used for proximity sensing, speed detection, RPM measurement

INTERFACE Digital (GPIO)

USD $1 – $2

2. REED SWITCH (Normally Open / NC)

Mechanical switch operated by magnetic field.

FEATURES

- Simple and reliable
- No power required
- Normally Open (NO) / Normally Closed (NC)
- Long life
- Used in door sensors, alarms, position detection

INTERFACE Digital (Switch)

USD $1 – $2

3. HMC5883L COMPASS MODULE (3-Axis Magnetometer)

3-axis digital compass for measuring magnetic field in X, Y, Z axes.

FEATURES

- 3-Axis measurement (X, Y, Z)
- High sensitivity (±8 gauss)
- I²C Interface (up to 400 kHz)
- Operating voltage: 2.16V – 3.6V
- Used in compasses, navigation, robotics, orientation sensing

INTERFACE I²C

PRICE (Approx.) USD $4 – $8

4. QMC5883L COMPASS MODULE (3-Axis Magnetometer)

High performance 3-axis digital magnetometer.

FEATURES

- 3-Axis measurement (X, Y, Z)
- Wide range (±8 Gauss)
- High resolution (16-bit)
- I²C Interface (up to 400 kHz)
- Operating voltage: 2.7V – 3.6V
- Low power consumption
- Used in drones, GPS, robotics, navigation systems

INTERFACE I²C

PRICE (Approx.) USD $4 – $7

5. GIANT MAGNETORESISTIVE (GMR) SENSOR (AMC7120 / NVE Series)

Detects extremely small magnetic fields using GMR technology.

FEATURES

- Very high sensitivity
- Wide supply range (2.7V – 5.5V)
- Analog output (Proportional)
- Low noise, high stability
- Used in precision current sensing, linear position sensing, industrial and automotive applications

INTERFACE Analog Output

PRICE (Approx.) USD $3 – $10

COMMON APPLICATIONS

- RPM & Speed Measurement
- Position & Proximity Detection
- Compass & Navigation
- Robotics & Automation
- Security & Alarm Systems
- Automotive Systems

HOW THEY WORK

Magnetic Field → Sensor Detects Field Change → Converts to Electrical Signal → Processed by Microcontroller

TIPS FOR USE

- ✔ Keep sensors away from strong magnets & motors.
- ✔ Use proper power supply & decoupling capacitors.
- ✔ Calibrate compass sensors for accurate readings.
- ✔ Follow datasheet guidelines for best performance.

💡 Magnetic sensors are widely used in modern electronics for non-contact sensing, precision measurement and smart automation.

GAS / AIR QUALITY SENSORS

Detect harmful gases, VOCs and pollutants for a safer and healthier environment

Better Air Quality | Health Protection | Energy Efficiency | Data Driven Monitoring | Smart & Green Environment

MQ-2
Smoke, LPG, Propane, Hydrogen, Methane

FEATURES
- Detects LPG, Smoke, Alcohol, Propane, Hydrogen, Methane
- High sensitivity and fast response
- Adjustable sensitivity (potentiometer)
- Wide detection range
- Long life and low cost

DETECTION RANGE
200 – 10,000 ppm

INTERFACE
Analog (AO) / Digital (DO)

APPROX. PRICE
USD $2 – $5

MQ-3
Alcohol, Ethanol

FEATURES
- High sensitivity to Alcohol (Ethanol)
- Good stability and long life
- Fast response and quick recovery
- Adjustable sensitivity
- Simple drive circuit

DETECTION RANGE
0.05 – 10 mg/L (Alcohol)

INTERFACE
Analog (AO) / Digital (DO)

APPROX. PRICE
USD $2 – $5

MQ-4
Methane, CNG

FEATURES
- High sensitivity to Methane (CH_4) and CNG
- Fast response
- Stable performance
- Adjustable sensitivity
- Low power consumption

DETECTION RANGE
200 – 10,000 ppm (CH_4)

INTERFACE
Analog (AO) / Digital (DO)

APPROX. PRICE
USD $2 – $5

MQ-7
Carbon Monoxide (CO)

FEATURES
- High sensitivity to CO
- Good selectivity and stability
- Fast response and high reliability
- Simple drive circuit
- Long life

DETECTION RANGE
20 – 2,000 ppm (CO)

INTERFACE
Analog (AO) / Digital (DO)

APPROX. PRICE
USD $2 – $5

MQ-135
Air Quality, NH_3, NOx, Benzene, Smoke, CO_2

FEATURES
- Detects NH_3, NOx, Benzene, Smoke, CO_2 and other gases
- High sensitivity and wide detection
- Fast response
- Good stability and long life
- Adjustable sensitivity

DETECTION RANGE
10 – 10,000 ppm

INTERFACE
Analog (AO) / Digital (DO)

APPROX. PRICE
USD $2 – $5

CCS811
eCO_2, TVOC, Air Quality

FEATURES
- Measures eCO_2 (Equivalent CO_2) and TVOC
- Digital output (I²C)
- Low power consumption
- Built-in temperature and humidity compensation
- Ideal for IAQ monitoring

DETECTION RANGE
eCO_2: 400 – 8,192 ppm
TVOC: 0 – 1,188 ppb

INTERFACE
I²C (3.3V / 5V compatible)

APPROX. PRICE
USD $8 – $15

SGP30
VOC, Air Quality

FEATURES
- Detects VOC (Volatile Organic Compounds)
- Digital output (I²C)
- Low power consumption
- Built-in compensation for humidity and temperature
- Great for smart homes, HVAC, and monitoring

DETECTION RANGE
0 – 60,000 ppb VOC

INTERFACE
I²C (3.3V / 5V compatible)

APPROX. PRICE
USD $10 – $20

BME680
Gas, Pressure, Temp., Humidity (Environmental)

FEATURES
- Measures Gas, Pressure, Temperature and Humidity
- Detects VOC, CO_2 equivalent and air quality
- High accuracy and stability
- Low power consumption
- Smart indoor air quality monitoring

DETECTION RANGE
VOC: 0 – 60,000 ppb
Pressure: 300 – 1,100 hPa

INTERFACE
I²C / SPI (3.3V)

APPROX. PRICE
USD $10 – $18

TYPICAL APPLICATIONS

Air Quality Monitoring | Smart Home Automation | Industrial Safety | HVAC Systems | Environmental Monitoring | IoT & Smart Devices

TIPS FOR BEST RESULTS

- Calibrate sensors in clean air for accurate readings.
- Keep sensors away from water, alcohol vapors and high humidity.
- Allow warm-up time (24–48 hours) for MQ sensors for stable output.
- Use proper filtering and placement for reliable measurements.

NOTE

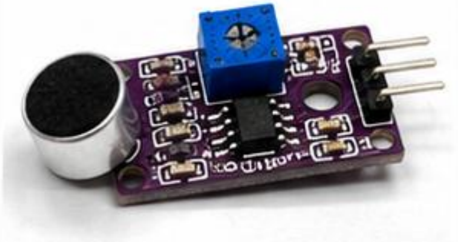
MQ series sensors are analog and need calibration. CCS811, SGP30 and BME680 are digital sensors with better accuracy and long term stability.

SOUND SENSORS

DETECT • MEASURE • LISTEN

Sound sensors convert sound waves into electrical signals that can be measured and processed by microcontrollers like Arduino or ESP32.

1. KY-037 SOUND SENSOR MODULE



FEATURES
- Uses LM393 comparator
- Digital output (DO) and Analog output (AO)
- Adjustable sensitivity (via potentiometer)
- Operating voltage: 3.3V – 5V DC
- Detects sound intensity level
- Onboard indicator LED
- Good for simple sound detection

Voltage 3.3V – 5V | Output DO, AO | Price $1 – $2

2. KY-038 MICROPHONE SENSOR MODULE

FEATURES
- Uses LM993 comparator
- Digital (DO) and Analog (AO) outputs
- Sensitivity adjustable
- Operating voltage: 3.3V – 5V DC
- Detects claps, knocks, loud sounds
- Compact and easy to use
- Indicator LED for output status

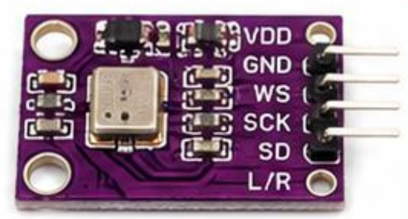
Voltage 3.3V – 5V | Output DO, AO | Price $1 – $2

3. MAX9814 MICROPHONE AMPLIFIER

FEATURES
- High sensitivity electret microphone amp
- Automatic Gain Control (AGC)
- Adjustable gain via potentiometer
- Wide supply range: 2.7V – 5.5V DC
- Low noise and high SNR
- Analog output (envelope signal)
- Ideal for audio level detection

Voltage 2.7V – 5.5V | Output Analog | Price $4 – $8

4. INMP441 I²S MEMS MICROPHONE



FEATURES
- Digital MEMS microphone
- I²S digital output interface
- High SNR and low power
- Wide supply range: 1.8V – 3.6V
- Omni-directional microphone
- Excellent for audio recording
- Works with ESP32, Arduino, Raspberry Pi

Voltage 1.8V – 3.6V | Output I²S Digital | Price $5 – $10

5. SPH0645LM4H-B MEMS MICROPHONE

FEATURES
- Digital MEMS microphone (Knowles SPH0645)
- I²S digital output
- Supply voltage: 1.8V – 3.6V
- High Acoustic Overloter Point (AOP)
- High SNR and wide dynamic range
- Omni-directional
- Ideal for audio capture and voice projects

Voltage 1.8V – 3.6V | Output I²S Digital | Price $5 – $10

COMMON APPLICATIONS

- Sound Level Meter
- Audio Reactive Projects
- Clap / Knock Detection
- Noise Alarm System
- Voice Recording
- Voice / Sound Controlled Robots

COMPARISON TABLE

Module	Type	Output Type	Sensitivity	Best For	Interface
KY-037	Electret Mic + Comparator	Digital, Analog	Adjustable	Simple sound detection	DO, AO
KY-038	Electret Mic + Comparator	Digital, Analog	Adjustable	Clap / knock detection	DO, AO
MAX9814	Mic Amplifier with AGC	Analog	High (AGC)	Sound level measurement	Analog
INMP441	Digital MEMS Mic	I²S Digital	High	Audio recording, ESP32	I²S
SPH0645	Digital MEMS Mic	I²S Digital	High	High quality audio capture	I²S

TIPS

- Use Analog output for level measurement.
- Use Digital output for simple sound detection.
- Adjust the potentiometer to set desired sensitivity.
- Place the microphone away from wind and vibration.
- For high quality audio use I²S microphones.

NOTE: Prices are approximate and may vary by supplier and location.

Check your local electronics store or online marketplaces for latest prices.

PERFECT FOR ARDUINO, ESP32, RASPBERRY PI & MORE!

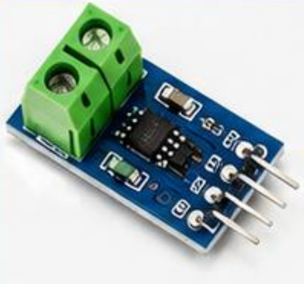
ELECTRICAL SENSORS
TYPES, FEATURES & MODULES

MEASURE ELECTRICAL PARAMETERS

Electrical sensors detect electrical parameters such as voltage, current, power and energy and convert them into signals that can be read by microcontrollers like Arduino / ESP32.

ACCURATE MEASUREMENT | PROTECTION & SAFETY | EASY TO USE MODULES | WIDE RANGE OF APPLICATIONS

ACS712 CURRENT SENSOR



Features
- Hall effect based
- Isolated sensing
- High accuracy
- Low power
- Fast response

Current Range: ±5A / ±20A / ±30A | Output: Analog

INA219 CURRENT / VOLTAGE SENSOR

Features
- Measures Voltage, Current & Power
- High accuracy
- I²C Interface
- Programmable Calibration

Voltage Range: 0 – 26V | Current Range: ±3.2A

INA226 CURRENT / VOLTAGE SENSOR

Features
- High precision
- Measures Voltage, Current, Power
- I²C Interface
- 16-bit ADC
- Wide range

Voltage Range: 0 – 36V | Current Range: ±5A

ZMPT101B AC VOLTAGE SENSOR

Features
- AC Voltage Measurement
- Galvanic Isolation
- Wide Voltage Range
- Easy Interface

Voltage Range: 80 – 260V AC | Output: Analog

SCT-013 CURRENT CLAMP SENSOR

Features
- Non-intrusive Current Sensing
- Easy to Use
- Safe & Isolated
- Compact Design

Current Range: 0 – 100A AC | Output: Analog

PZEM-004T ENERGY METER MODULE

Features
- Measures Voltage, Current, Power, Energy, Frequency
- High Accuracy
- TTL Serial Output
- Compact Size

Voltage: 80 – 260V AC | Current: 0 – 100A AC | Power: 0 – 22kW | Energy: 0 – 99999 kWh

ZMCT103C CURRENT TRANSFORMER MODULE

Features
- AC Current Measurement
- Galvanic Isolation
- High Linearity
- Good Accuracy

Current Range: 0 – 5A AC | Output: Analog

VOLTAGE DIVIDER MODULE

Features
- Scales down High Voltage
- Adjustable Ratio
- Simple & Low Cost
- Wide Applications

Input Voltage: Up to 25V DC | Output: Analog

HALL EFFECT CURRENT SENSOR (ACS758)

Features
- Hall effect based
- Galvanic Isolation
- Low Offset
- High Accuracy
- High Bandwidth

Current Range: ±50A / ±100A / ±150A | Output: Analog

WATT METER MODULE (HLW8012)

Features
- Measures Voltage, Current, Power, Energy
- Low Power
- Compact
- Easy to Interface

Voltage: 90 – 250V AC | Power: 0 – 3.6kW

APPLICATIONS

Energy Monitoring | Power Measurement | Industrial Automation | Home Automation | Renewable Energy | Battery Management

INTERFACES

Analog Output | I²C Interface | UART / TTL

WORKS WITH

ARDUINO | ESP32 | Raspberry Pi | and more...

FORCE / WEIGHT SENSORS

Measure Force, Pressure, Weight, Load, Strain & Human Touch

- **High Accuracy** — Precise & Reliable Measurements
- **Wide Range** — From grams to tons
- **Easy Interface** — Works with Arduino ESP32, Raspberry Pi
- **Built for Projects** — Robotics, IoT, Industrial & DIY Applications

COMMON FEATURES

- Detect force, pressure or load and convert to electrical signal
- High sensitivity and good linearity
- Low power consumption
- Easy to interface (Analog / Digital via HX711, etc.)
- Wide measurement range options
- Compact and easy to integrate

1. FSR402 FORCE SENSOR

- **Type** : Force Sensitive Resistor
- **Range** : 0.1 N – 10 N (approx.)
- **Output** : Resistance change (Analog)
- **Features:**
  - Thin, Flexible & Lightweight
  - Detects applied force or pressure
  - Very low cost & easy to use
- **Applications:**
  - Touch detection, Robotics, Wearables, Pressure sensing, Human interface

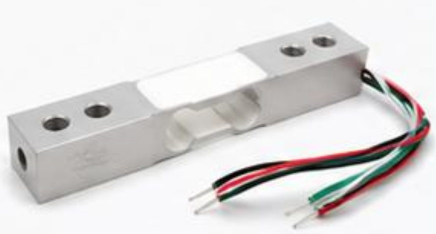
Price: USD \$6 – \$12

2. HX711 LOAD CELL AMPLIFIER KIT

- **Type** : 24-bit Load Cell Amplifier
- **Supply** : 2.6 – 5.5V
- **Interface:** Digital (DT, SCK)
- **Features:**
  - High precision 24-bit ADC
  - Works with all 4/6-wire load cells
  - Built-in low noise programmable gain
  - Easy to use with Arduino/ESP32
- **Applications:**
  - Weighing scales, Industrial systems, Load measurement

Price: USD \$4 – \$10

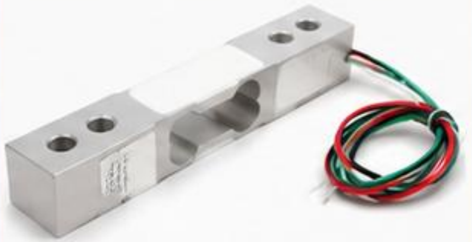
3. 5KG LOAD CELL



- **Type** : Strain Gauge Load Cell
- **Capacity** : 5 kg
- **Output** : ~1.0 – 2.0 mV/V
- **Features:**
  - High accuracy & good stability
  - Widely used with HX711 module
  - Aluminum construction
- **Applications:**
  - Weighing scales, Hoppers, DIY balance, Inventory systems

Price: USD \$5 – \$10

4. 20KG LOAD CELL



- **Type** : Strain Gauge Load Cell
- **Capacity:** 20 kg
- **Output** : ~1.0 – 2.0 mV/V
- **Features:**
  - Higher load capacity
  - High accuracy & durability
  - Compatible with HX711 module
- **Applications:**
  - Platform scales, Hopper weighing, Industrial load measurement

Price: USD \$8 – \$15

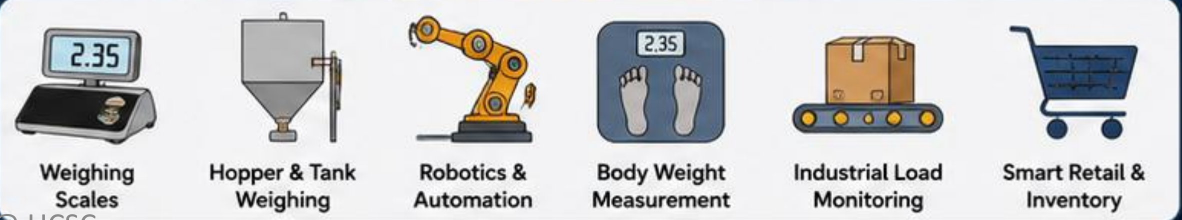
5. 50KG LOAD CELL

- **Type** : Strain Gauge Load Cell
- **Capacity:** 50 kg
- **Output** : ~1.0 – 2.0 mV/V
- **Features:**
  - Heavy duty & high reliability
  - Excellent for industrial use
  - Works with HX711 amplifier
- **Applications:**
  - Floor scales, Tank weighing, Industrial automation

Price: USD \$10 – \$18

HOW IT WORKS

COMMON APPLICATIONS



Tip: For best results, use a quality HX711 module and calibrate your system with known weights.

Note: Do not exceed the maximum load capacity of the sensor.

WATER / LIQUID SENSORS

FOR ARDUINO / ESP32 / IOT PROJECTS

WATER MONITORING | AGRICULTURE | INDUSTRIAL AUTOMATION | SMART HOME

1. FLOAT SWITCH

- Detects liquid level by float movement
- Simple ON/OFF operation
- No moving parts in contact with electronics
- Working Voltage: 3V – 24V DC
- Max Current: 0.5A
- Easy to install

USD $2 – $5

2. WATER LEVEL SENSOR

- Detects presence of water
- Analog output
- Comparators for digital output (D0 / A0)
- Operating Voltage: 3.3V – 5V
- LED indicator
- Low cost and easy to use

USD $2 – $4

3. YF-S201 FLOW SENSOR

- Measures water flow rate
- Working Voltage: 5V DC
- Flow Range: 1 – 30 L/min
- Output: Frequency Pulse
- Accuracy: ±10%
- Withstands water pressure up to 1.75 MPa

USD $5 – $8

4. CAPACITIVE WATER LEVEL SENSOR

- Detects liquid through container (non-contact)
- Capacitive sensing technology
- Operating Voltage: 3.3V – 5V
- Adjustable sensitivity
- Corrosion resistant
- Analog output

USD $4 – $8

5. pH METER MODULE

- Measures acidity / alkalinity
- pH Range: 0 – 14
- Working Voltage: 5V DC
- Analog output (BNC probe)
- High accuracy
- Used in water quality monitoring, hydroponics, aquariums

USD $15 – $35

6. TDS SENSOR MODULE

- Measures Total Dissolved Solids
- TDS Range: 0 – 1000 ppm
- Working Voltage: 3.3V – 5V
- Analog output
- Used for water purity testing, RO systems, aquariums

USD $15 – $30

APPLICATIONS

- Water Tanks & Reservoirs
- Water Quality Monitoring
- Aquarium Monitoring
- Hydroponics & Irrigation
- Industrial Process Control
- Leak Detection & Alert Systems
- Smart Home Automation

QUICK COMPARISON

Sensor	Detects / Measures	Output	Voltage	Key Feature	Price (USD)
Float Switch	Liquid Level (ON/OFF)	Digital (ON/OFF)	3V – 24V DC	Simple & Reliable	$2 – $5
Water Level Sensor	Water Presence / Level	Analog / Digital	3.3V – 5V	Low Cost, Easy to Use	$2 – $4
YF-S201 Flow Sensor	Flow Rate	Pulse (Frequency)	5V DC	Measures Flow in L/min	$5 – $8
Capacitive Level Sensor	Liquid Level (Non-contact)	Analog	3.3V – 5V	Detects Through Container	$4 – $8
pH Meter Module	Acidity / Alkalinity	Analog	5V DC	pH 0 – 14, High Accuracy	$15 – $35
TDS Sensor Module	Dissolved Solids (ppm)	Analog	3.3V – 5V	Tests Water Purity	$15 – $30

TIPS FOR USE

- Calibrate pH and TDS sensors regularly for accurate readings.
- Keep sensors clean and free from deposits.
- Use proper power supply and avoid reverse polarity.
- Ensure proper sealing for outdoor / wet environments.
- Use shielding or twisted wires for long cable lengths.

★ These sensors help you build smart and efficient water monitoring systems for a better tomorrow. ★

POSITION SENSORS

Types, Features & Applications

Position sensors detect the position, angle or displacement of an object and convert it into an electrical signal.

Applications

- Robotics & Automation
- CNC Machines
- Industrial Control
- Mechatronics

1. POTENTIOMETER

Features

- Simple & Low cost
- Provides Analog output
- Contact based
- 10Ω to 1MΩ resistance range

Output: Analog (Voltage Divider) | **Interface:** Analog

Applications: Volume Control, Joysticks, Position Feedback

2. ROTARY ENCODER

Features

- Detects rotation & direction
- Incremental output
- High accuracy
- Digital output
- Optical / Mechanical types

Output: Digital (Pulse) | **Interface:** Digital (GPIO)

Applications: Motor Control, CNC, Robotics, User Interface

3. LINEAR POTENTIOMETER

Features

- Measures linear displacement
- Analog output
- Smooth & simple
- Various stroke lengths available

Output: Analog (Voltage Divider) | **Interface:** Analog

Applications: Actuator Position, Industrial Automation

4. LINEAR ENCODER (OPTICAL)

Features

- High precision
- Non-contact type
- High resolution
- Long measuring length
- Digital output

Output: Digital (Quadrature / SSI) | **Interface:** Digital / SSI

Applications: CNC Machines, 3D Printers, Precision Systems

5. HALL EFFECT POSITION SENSOR

Features

- Magnetic field based
- Contactless
- High reliability
- Detects linear or angular position
- Digital / Analog output

Output: Digital / Analog | **Interface:** Analog / Digital

Applications: Motor Commutation, Speed Detection, Position Sensing

6. RESOLVER

Features

- Rugged & reliable
- High accuracy
- Works in harsh environment
- Provides absolute position
- Analog output

Output: Analog (Sin/Cos) | **Interface:** Analog

Applications: Servo Motors, Aerospace, Industrial Drives

7. LVDT (LINEAR VARIABLE DIFFERENTIAL TRANSFORMER)

Features

- Very high accuracy
- Excellent linearity
- No electrical contact
- High reliability
- Measures linear displacement

Output: Analog | **Interface:** Analog

Applications: Industrial Measurement, Test Equipment, Position Feedback

8. MAGNETIC ENCODER (ABSOLUTE)

Features

- Absolute position
- High resolution
- Magnetic sensing
- Contactless
- Resistant to dust, oil & vibration

Output: Digital (Absolute) | **Interface:** SPI / I²C / SSI

Applications: Robotics, Servo Systems, Industrial Control

KEY POINTS

- Position sensors are used to detect linear or angular position.
- They can be contact or non-contact based.
- Output can be Analog, Digital, or Pulse based.
- Selection depends on accuracy, environment & application.

COMMON INTERFACES

Analog | Digital (GPIO) | I²C | SPI | SSI

Learn More
Scan QR Code

IMAGE SENSORS – MODULES & FEATURES

POPULAR IMAGE SENSOR MODULES FOR ARDUINO, ESP32 & EMBEDDED SYSTEMS

- ✓ Capture Images & Video
- ✓ High Sensitivity
- ✓ Low Power Options
- ✓ Multiple Interfaces

COMMON INTERFACES
- DVP (Parallel)
- SPI
- I2C (for control)
- USB / MIPI CSI (on SBC)

OV7670 Camera Module

KEY FEATURES
- VGA Resolution (640 x 480)
- Supports RGB, YUV Output
- Built-in Image Processing
- Adjustable: Brightness, Contrast, Saturation, White Balance

RESOLUTION	VGA (640 x 480)
INTERFACE	DVP (Parallel)
SUPPLY VOLTAGE	3.0V ~ 3.6V
VIEW ANGLE	~ 60°

TYPICAL USES: DIY Projects · Robotics · Surveillance

OV2640 Camera Module

KEY FEATURES
- 2 MP Resolution (1600 x 1200)
- Supports UXGA, SXGA, VGA, QVGA
- JPEG Compression Supported
- Auto Exposure, AWB, AEC, AGC

RESOLUTION	2 MP (1600 x 1200)
INTERFACE	DVP (Parallel)
SUPPLY VOLTAGE	2.5V ~ 3.0V
VIEW ANGLE	~ 65°

TYPICAL USES: IoT Devices · ESP32-CAM · Home Security

ESP32-CAM (OV2640)

KEY FEATURES
- Built-in ESP32 + OV2640 Camera
- Wi-Fi + Bluetooth Support
- MicroSD Card Slot
- Ideal for IoT & Wireless Projects

RESOLUTION	2 MP (1600 x 1200)
INTERFACE	DVP (Parallel)
SUPPLY VOLTAGE	5V (Via USB)
VIEW ANGLE	~ 66°

TYPICAL USES: IoT Camera · Wireless Video · Remote Monitoring

Raspberry Pi Camera Module v3

KEY FEATURES
- 12 MP Resolution (4056 x 3040)
- Sony IMX708 Sensor
- High Dynamic Range
- Supports 1080p & 720p Video

RESOLUTION	12 MP (4066 x 3040)
INTERFACE	MIPI CSI
SUPPLY VOLTAGE	3.3V
VIEW ANGLE	~ 75°

TYPICAL USES: High Quality Imaging · Computer Vision · Research & AI

Arducam OV5647

KEY FEATURES
- 5 MP Resolution (2592 x 1944)
- Wide 69° Field of View
- Supports JPEG / RAW Output
- Works with SPI Interface

RESOLUTION	5 MP (2592 x 1944)
INTERFACE	SPI
SUPPLY VOLTAGE	3.3V
VIEW ANGLE	~ 69°

TYPICAL USES: Raspberry Pi · Embedded Vision · AI Projects

USB Camera Module (UVC)

KEY FEATURES
- USB Video Class (UVC) Compliant
- No Driver Required (Plug & Play)
- 720p / 1080p Options
- Compatible with PC & SBC

RESOLUTION	720p / 1080p
INTERFACE	USB 2.0
SUPPLY VOLTAGE	5V (Via USB)
VIEW ANGLE	~ 60° – 90°

TYPICAL USES: Video Conferencing · Streaming · PC Vision Apps

QUICK COMPARISON

RESOLUTION	Higher is better	★★	★★½	★★★	★★★★½	★★★★★	★★★½	★★★★
IMAGE QUALITY	(Low Light Performance)	★★½	★★★½	★★★½	★★★★★	★★★½	★★½	★★★★
EASE OF USE	(For Beginprox.) (USD)	$5 - $8	$5 - $10	$8 - $15	$8 - $35	$25 - $35	$10 - $20	$15 - $30

TIPS
- For IoT & wireless projects → ESP32-CAM
- For high quality stills → Raspberry Pi Camera v3
- For general purpose → OV2640
- For easy USB connection → UVC Camera Module

Scan for Datasheets & More Info

BIOMETRIC SENSORS

Smart Sensing for Secure Identification

1 FINGERPRINT SENSOR
R307 Optical Fingerprint Module

FEATURES
- Operating Voltage: 3.3V
- Fingerprint storage: 1000
- Interface: UART (TTL)
- Fast recognition
- Low power consumption
- High security

Approx. $15 – $30

APPLICATIONS: Access control, Attendance systems, Door locks, Safety systems

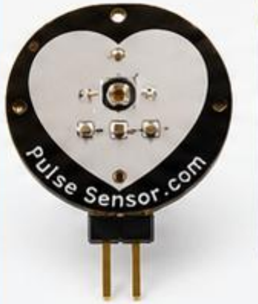
2 FINGERPRINT SENSOR
AS608 Optical Fingerprint Module

FEATURES
- Operating Voltage: 3.6 – 6.0V
- Fingerprint storage: 1000
- Interface: UART (TTL)
- High speed recognition
- Built-in fingerprint algorithm
- Small size & easy to use

Approx. $15 – $25

APPLICATIONS: Access control, Time & attendance, Smart door locks, Identifcations

3 PULSE SENSOR
Pulse Sensor Amped



FEATURES
- Operating Voltage: 3.3 – 5V
- Measures heart rate (BPM)
- Easy to use analog output
- Low power consumption
- Works with Arduino, ESP32, etc.

Approx. $5 – $8

APPLICATIONS: Fitness tracking, Heart rate monitoring, Health & wellness devices

4 HEART RATE & OXYGEN SENSOR
MAX30102 Module

FEATURES
- Operating Voltage: 1.8 – 3.3V
- Measures Heart Rate & SpO_2 (Blood Oxygen)
- I²C Interface
- Low power, high sensitivity
- Built-in LED, photodiode, optics & low noise electronics

Approx. $5 – $10

APPLICATIONS: Wearables, Fitness bands, Medical monitoring, IoT health devices

5 IRIS RECOGNITION SENSOR
WISE-Iris-G2

FEATURES
- Operating Voltage: 5V
- Recognition distance: 30 – 80 cm
- High accuracy
- Fast recognition (<1 sec)
- UART / USB Interface
- High security

Approx. $70 – $120

APPLICATIONS: High security access control, Government systems, Research applications

6 FACE RECOGNITION CAMERA
ESP32-CAM AI Module

FEATURES
- Operating Voltage: 5V
- Camera: OV2640 (2MP)
- Supports face detection & recognition (AI)
- Wi-Fi + Bluetooth (ESP32)
- Compact size

Approx. $8 – $15

APPLICATIONS: Smart door locks, Attendance systems, Surveillance, IoT applications

7 PALM VEIN SCANNER
R503 Palm Vein Module

FEATURES
- Operating Voltage: 5V
- Palm vein recognition
- High accuracy & security
- USB / UART Interface
- Anti-spoofing (liveness detection)

Approx. $120 – $180

APPLICATIONS: Banking, High security access, Time attendance, Secure authentication

8 VOICE RECOGNITION MODULE
LD3320

FEATURES
- Operating Voltage: 3.3 – 5V
- Recognizes voice commands
- Up to 80 voice commands
- UART Interface
- Easy to train & use

Approx. $6 – $12

APPLICATIONS: Voice controlled devices, Home automation, Security systems, Robotics

WHY USE BIOMETRIC SENSORS?
- Unique to individuals
- High accuracy & reliability
- Improved security
- Convenient & touchless options
- Wide range of applications

COMMON INTERFACES
UART | I²C | SPI | USB | Analog

NOTE
Prices are approximate and may vary based on brand, quality and location. (USD – May 2025)

ACTUATORS – VISUAL

Display and Light Output Devices for Arduino, ESP32 & Embedded Projects

Used to give visual feedback, indication and display information in embedded systems.

1 5 mm LED

FEATURES

- Low power consumption
- Forward voltage: 1.8 – 3.3V
- Current: 10 – 20 mA
- Wide viewing angle
- Used as simple indicator lights

2 RGB LED

FEATURES

- Can display millions of colors
- Common Anode / Common Cathode
- Forward voltage: 2.0 – 3.4V (per color)
- Current: 20 mA per color
- Controlled by PWM or digital pins

3 WS2812 RGB LED (Addressable)

FEATURES

- Individually addressable LEDs
- Built-in driver IC (WS2812B)
- 5V operation, single data line
- 16.7 million colors
- Ideal for strips, matrices, rings

4 NeoPixel Ring (WS2812)

FEATURES

- Pre-arranged addressable LEDs
- Easy to use with one data pin
- 5V operation
- Bright, vibrant colors
- Available in various sizes (12, 16, 24, 32 LEDs ...)

5 16x2 LCD Display

FEATURES

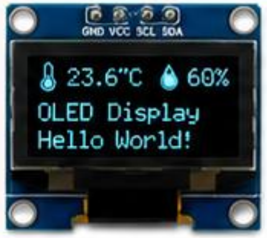
- 16 characters x 2 lines
- HD44780 compatible
- 5V operation
- I2C backpack available (SDA, SCL)
- Used to display text and numbers

6 20x4 LCD Display

FEATURES

- 20 characters x 4 lines
- HD44780 compatible
- 5V operation
- I2C backpack supported
- More space for displaying data

7 OLED Display (SSD1306)



FEATURES

- 0.96", 128x64 pixels (common)
- I2C interface (SDA, SCL)
- Self-emissive: No backlight needed
- High contrast, wide viewing angle
- Very low power consumption

8 TFT Display (ILI9341)

FEATURES

- 2.2", 2.4", 2.8", 3.2" sizes available
- Resolution: 240x320 (2.4" typical)
- SPI interface
- 65K / 262K colors
- Good for graphics, images & UI

9 7-Segment Display

FEATURES

- Numeric display (0–9)
- Common Cathode / Anode
- Size: 0.36", 0.56", 0.8" etc.
- Controlled by digital pins or through driver (74HC595, MAX7219)
- Used for counters, timers, clocks

WHY USE VISUAL ACTUATORS?

- Provide user feedback
- Indicate system status
- Display sensor data
- Enhance user interface
- Improve user experience
- Essential for debugging

COMMON INTERFACES

- Digital I/O (LED, 7-Segment)
- PWM (LED Brightness)
- I2C (LCD, OLED)
- SPI (TFT, Displays)

APPLICATIONS

- Status Indicators
- User Interfaces
- Data Display
- Alarms & Alerts
- Decorative Lighting

Audio Actuators for Arduino / ESP32

Convert electrical signals into sound for alerts, alarms, music, voice and notifications.

Active Buzzer

$1–2

Example: KY-012 / Passive Active Buzzer

- Built-in oscillator
- Produces continuous tone when powered
- Simple to use (ON/OFF)
- Voltage: 3.3V – 12V
- Current: ~15–30 mA

Best for: Simple alarms, beeps, indicators

Passive Buzzer

$1–2

- No built-in oscillator
- Requires PWM / square wave from MCU
- Can play different tones
- Voltage: 3.3V – 12V
- Current: ~10–25 mA

Best for: Custom beeps, tunes, melody

Mini Speaker (Driver)

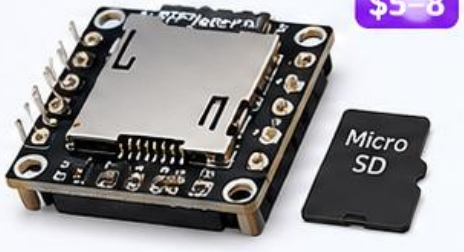
$2–5

- Louder than buzzers
- Good frequency response (200 Hz – 20 kHz)
- Impedance: 4Ω / 8Ω
- Power: 0.5W – 3W
- Needs amplifier for best sound

Best for: Announcements, alerts, voice prompts

DFPlayer Mini MP3 Module

$5–8

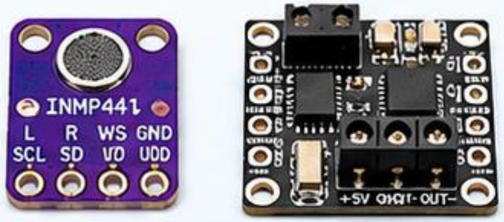


- Plays MP3 / WAV files from microSD card
- Communication: UART (TX/RX)
- Supply Voltage: 3.2V – 5V
- Output Power: ~3W (with small speaker)
- Easy commands via Serial

Best for: Voice messages, music playback, effects

I²S Digital MEMS Speaker (INMP441 / SPH0645 + Amp)

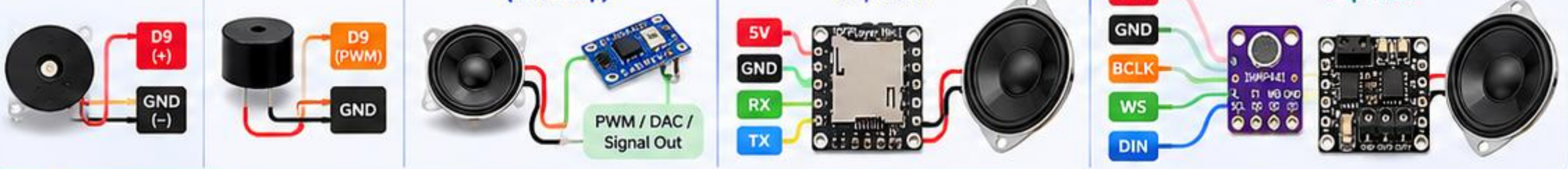
$5–10



- High quality digital audio
- I²S interface (digital)
- Clearer voice and music
- Low noise, high fidelity
- Needs I²S mic + Class-D Amp (e.g., PAM8403, MAX98357A)

Best for: High-fidelity audio, recording & smart devices

How to Connect (Typical)



Common Applications

- **Alarms & Alerts** (doorbell, security, fire)
- **Notifications** (Push alerts, reminders)
- **Music & Melody** (Tunes, chimes, songs)
- **Voice Prompts** (Announcements, instructions)
- **IoT Devices** (Smart home, wearables, robots)

Quick Comparison

Type	Loudness	Tone Control	Ease of Use	Best For
Active Buzzer	★☆☆	★☆☆	★★★★★	Simple beeps
Passive Buzzer	★★☆	★★★☆	★★★★☆	Custom tones
Mini Speaker	★★★★☆	★★★★☆	★★★☆☆	Louder sound
DFPlayer Mini	★★★★☆	★★★★★	★★★★☆	MP3 / Voice
I²S Digital Audio	★★★★★	★★★★★	★★★☆☆	High quality

Pro Tips

- Use PWM pins for Passive Buzzers to create different notes.
- Always match speaker impedance with amplifier output.
- For DFPlayer, use a quality microSD card (FAT32, 8–32GB).

Popular Amplifier ICs (for Speakers)

- PAM8403 → 3W Stereo Class-D
- MAX98357A → High efficiency
- LM386 → Simple audio amplifier

MOTION ACTUATORS

Types, Features & Applications for Arduino / ESP32 / Embedded Projects

1. SG90 SERVO MOTOR

FEATURES
- Operating Voltage: 4.8 – 6V
- Stall Torque: ~1.8 kg·cm (4.8V)
- Rotation: 0° – 180°
- Control: PWM (50 Hz)
- Lightweight & compact
- Plastic gears

TYPICAL USES

Robotics, RC toys, camera mounts, mini mechanisms

2. MG996R SERVO MOTOR

FEATURES
- Operating Voltage: 4.8 – 7.2V
- Stall Torque: ~10 kg·cm (6V)
- Rotation: 0° – 180°
- Control: PWM (50 Hz)
- All metal gears
- High torque & durability

TYPICAL USES

Robot arms, pan-tilt, mechanical grippers, RC cars/planes

3. NEMA17 STEPPER MOTOR

FEATURES
- Step Angle: 1.8° (200 steps/rev)
- Voltage: 12 – 24V
- Current: 1.2 – 1.7A (typ.)
- High precision positioning
- Excellent holding torque
- Easy open-loop control

TYPICAL USES

3D Printers, CNC machines, plotters, automation

4. 28BYJ-48 STEPPER MOTOR

FEATURES
- Step Angle: 5.625° (64 steps/rev)
- Voltage: 5V
- Current: ~240mA
- Comes with ULN2003 driver
- Low cost & easy to use
- Good for light load

TYPICAL USES

DIY projects, camera sliders, small position control

5. TT GEARED MOTOR (DC)

FEATURES
- Operating Voltage: 3 – 12V
- Gear Ratio: 48:1 (typ.)
- No-load Speed: ~150 RPM (6V)
- High torque
- Compact & lightweight
- Easy mounting

TYPICAL USES

Robot cars, line followers, small automation projects

6. 12V DC GEAR MOTOR

FEATURES
- Operating Voltage: 6 – 12V
- Gear Ratio: 100:1 (typ.)
- No-load Speed: ~60 RPM (12V)
- High torque & efficiency
- Metal gear construction
- Reversible direction

TYPICAL USES

Robots, conveyors, door locks, motorized applications

7. 775 DC MOTOR

FEATURES
- Operating Voltage: 12 – 24V
- No-load Speed: ~5000 – 10000 RPM
- High power & high speed
- Durable & versatile
- Easy to adapt with gearboxes

TYPICAL USES

RC cars, power tools, pumps, high speed applications

8. BLDC MOTOR (A2212)

FEATURES
- KV Rating: 2200KV (typ.)
- Voltage: 7.4 – 11.1V (2S – 3S LiPo)
- High efficiency & long life
- Smooth & quiet operation
- Requires ESC to drive
- High power-to-weight ratio

TYPICAL USES

Drones, RC aircraft, robotics, high performance projects

MOTOR DRIVERS / CONTROLLERS (OVERVIEW)

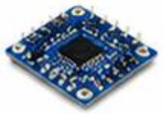
ULN2003

- For 28BYJ-48
- 5V operation
- Easy to use
- Low current
- ~$2 – $4

L298N

- Dual H-Bridge
- 5 – 35V
- Up to 2A
- Widely used
- ~$4 – $8

TB6612FNG



- Dual H-Bridge
- 2.5 – 13.5V
- Up to 1.2A
- Low heat
- ~$5 – $8

A4988

- Stepper Driver
- 8 – 35V
- Up to 2A
- Microstepping
- ~$3 – $6

DRV8825

- Stepper Driver
- 8.2 – 45V
- Up to 2.2A
- Quiet operation
- ~$4 – $8

TMC2209

- Stepper Driver
- 4.75 – 29V
- Up to 2A (RMS)
- UART control
- Very quiet
- ~$8 – $15

BTS7960

- High Power H-Bridge
- 6 – 27V
- Up to 43A
- For big motors
- ~$10 – $18

COMMON APPLICATIONS

Robot Arms

Mobile Robots

CNC Machines

3D Printers

RC Vehicles

Drones

Conveyors

Automation Systems

Note: Prices are approximate (USD) and may vary by region and supplier.

Choose the right actuator for the right motion!

Use proper driver, power supply & protection for best performance.

MOTOR DRIVERS COMPARISON GUIDE

★ — ★ Popular Motor Driver Modules for Arduino / ESP32 / Raspberry Pi Projects ★

★ HOW TO CHOOSE?

- For small DC motors → L9110, DRV8833
- For medium DC motors → L298N, TB6612FNG
- For high current DC motors → BTS7960
- For stepper motors → A4988, DRV8825, TMC2209

L9110

- Compact and low cost
- Good for small DC motors
- Built-in diodes
- Easy to use
- Low standby current

Motor Voltage	Peak Current	Channels	Control
2.5 – 12V	800mA	2	IN1, IN2

DRV8833

- Dual H-Bridge driver
- Low RDS(on) for efficiency
- Built-in current limiting
- Thermal shutdown
- Small size

Motor Voltage	Peak Current	Channels	Control
2.7 – 10.8V	1.5A	2	IN1, IN2

TB6612FNG

- Dual H-Bridge driver
- High efficiency
- Low standby current
- Built-in thermal shutdown
- Forward / Reverse / Brake

Motor Voltage	Peak Current	Channels	Control
2.5 – 13.5V	1.2A (3.2A Peak)	2	PWM / IN1, IN2

L293D

- Quadruple half-H driver
- Can drive DC motors and stepper motors
- Built-in back EMF diodes
- Very popular & reliable

Motor Voltage	Peak Current	Channels	Control
4.5 – 36V	600mA (per channel)	2	IN1~IN4

L298N

- Dual full-bridge driver
- High voltage & current
- Built-in heat sink
- Enable pin for speed control
- Widely used

Motor Voltage	Peak Current	Channels	Control
5 – 35V	2A (per channel)	2	ENA/ENB IN1~IN4

A4988

- Stepper motor driver
- Easy to use
- Microstepping up to 1/16
- Built-in current regulator
- Great for 3D printers and CNC

Motor Voltage	Current	Microstepping	Interface
8 – 35V	2A (max with cooling)	1, 1/2, 1/4, 1/8, 1/16	STEP/DIR

DRV8825

- Stepper motor driver
- Microstepping up to 1/32
- Low RDS(on)
- Built-in current regulation
- Silent & efficient

Motor Voltage	Current	Microstepping	Interface
8.2 – 45V	2.2A (max with cooling)	1, 1/2, 1/4, 1/8, 1/16, 1/32	STEP/DIR

TMC2209

- Ultra silent stepper driver
- UART communication
- Microstepping up to 1/256
- StallGuard (sensorless homing)
- CoolStep (auto current)
- Great for 3D printers

Motor Voltage	Current	Microstepping	Interface
4.75 – 29V	2.8A RMS (3.2A Peak)	1 – 256	STEP/DIR UART

BTS7960

- High current dual H-Bridge driver
- Perfect for large DC motors
- Very low RDS(on)
- Over current, over temperature protection

Motor Voltage	Peak Current	Channels	Control
6 – 27V	43A (per channel)	2	PWM / IN1, IN2

QUICK COMPARISON

Driver	Type	Motor Type	Max Current (Per Channel)	Motor Voltage
L9110	H-Bridge	DC	800mA	2.5 – 12V
DRV8833	H-Bridge	DC	1.5A	2.7 – 10.8V
TB6612FNG	H-Bridge	DC	1.2A (3.2A Peak)	2.5 – 13.5V
L293D	H-Bridge	DC / Stepper	600mA	4.5 – 36V
L298N	H-Bridge	DC / Stepper	2A	5 – 35V
A4988	Stepper	Stepper	2A	8 – 35V
DRV8825	Stepper	Stepper	2.2A	8.2 – 45V
TMC2209	Stepper	Stepper	2.8A RMS	4.75 – 29V
BTS7960	H-Bridge	DC	43A	6 – 27V

PROTECTION FEATURES
Most modules include protection against over-current, over-temperature and reverse voltage.

COOLING
For high current applications, use proper heat sinks and ensure good ventilation.

PWM CONTROL
Use PWM on ENA/ENB pins (or dedicated PWM pins) for speed control of DC motors.

NOTES
Always check the datasheet before use.
Do not exceed voltage or current ratings.

DC = DC Motor
Stepper = Stepper Motor
PWM = Pulse Width Modulation

RELAY MODULES

Switch Anything, Control Everything!

Safe • Reliable • Easy to Use

Home Automation | Industrial Control | DIY Projects | Safety & Protection

1 1-CHANNEL RELAY MODULE

FEATURES
- Controls 1 high-power appliance
- Opto-isolated for protection
- LED indicator (Power & Relay)
- Trigger: High/Low Level

Control Voltage: 5V DC
Load Voltage: 250V AC / 30V DC
Max Load Current: 10A
Size: 50 x 26 x 18 mm

BEST FOR: Simple switching, small DIY projects

2 2-CHANNEL RELAY MODULE

FEATURES
- Controls 2 devices independently
- Opto-isolated
- LED indicators
- Trigger: High/Low Level

Control Voltage: 5V DC
Load Voltage: 250V AC / 30V DC
Max Load Current: 10A (per channel)
Size: 50 x 43 x 18 mm

BEST FOR: Controlling two appliances, automation

3 4-CHANNEL RELAY MODULE

FEATURES
- Controls 4 devices independently
- Opto-isolated
- LED indicators
- Trigger: High/Low Level

Control Voltage: 5V DC
Load Voltage: 250V AC / 30V DC
Max Load Current: 10A (per channel)
Size: 75 x 55 x 20 mm

BEST FOR: Home automation, multi-device control

4 8-CHANNEL RELAY MODULE

FEATURES
- Controls 8 devices independently
- Opto-isolated
- LED indicators
- Trigger: High/Low Level

Control Voltage: 5V DC
Load Voltage: 250V AC / 30V DC
Max Load Current: 10A (per channel)
Size: 138 x 55 x 20 mm

BEST FOR: Large automation systems, equipment control

5 SOLID STATE RELAY (SSR) MODULE

FEATURES
- No moving parts
- Silent operation
- Fast switching
- Long life
- Opto-isolated

Control Voltage: 3–32V DC
Load Voltage: 24–380V AC
Max Load Current: 2A / 5A / 10A / 25A*
Size: 45 x 30 x 20 mm

BEST FOR: Silent switching, frequent ON/OFF cycles

6 HIGH POWER RELAY MODULE

FEATURES
- High current handling
- Suitable for demanding appliances
- Opto-isolated
- LED indicator

Control Voltage: 12V DC
Load Voltage: 250V AC / 30V DC
Max Load Current: 20A / 30A
Size: 60 x 31 x 25 mm

BEST FOR: High power loads, motors, heaters

7 LATCHING RELAY MODULE

FEATURES
- Maintains last state without power
- Bi-stable (Set / Reset)
- Low power consumption
- Opto-isolated

Control Voltage: 5V DC
Load Voltage: 250V AC / 30V DC
Max Load Current: 10A
Size: 52 x 34 x 18 mm

BEST FOR: Battery powered projects, energy saving

8 RELAY MODULE WITH OPTOCOUPLER

FEATURES
- Optocoupler isolation
- Protects microcontroller from high voltage
- Stable & reliable
- LED indicators

Control Voltage: 5V DC
Load Voltage: 250V AC / 30V DC
Max Load Current: 10A (per channel)
Size: Varies (1/2/4/8 Channel)

BEST FOR: Microcontroller protection, industrial use

COMMON FEATURES (ALL MODULES)

- Opto-isolation for safety
- Easy to use with Arduino / ESP32 / Raspberry Pi
- Screw terminals for secure connections
- LED indicators for status monitoring
- Overload & short circuit protection (Design dependent)

NOTE:

- Always use within the specified voltage and current limits.
- For inductive loads (motors, pumps), use snubber circuits if needed.
- Ensure proper ventilation for high current applications.

LEGEND

- Control Voltage (Logic Side)
- Load Voltage (High Power Side)
- Max Load Current (Per Channel)
- (L x W x Size

SOLENOIDS & VALVES

Types, Features & Applications

WHY SOLENOIDS & VALVES?

- Convert electrical energy to mechanical motion
- Fast response & high reliability
- Used in automation, robotics, irrigation, industrial control, appliances & more

Fast Response | High Precision | Reliable & Durable | Easy to Control | Wide Range of Applications

5V SOLENOID

- Compact size
- Pull type
- Low power consumption
- Quick response

Voltage: 5V DC | Price: $5 – $10

- Use: Locking, Latching, Small automation

12V SOLENOID

- Stronger force
- Push/Pull type
- Durable metal construction
- Easy mounting

Voltage: 12V DC | Price: $8 – $15

- Use: Automation, Vending machines, Door locks

SOLENOID DOOR LOCK

- Secure locking
- Holding force
- Fail-safe operation (optional)
- Compact & robust

Voltage: 12V DC | Price: $8 – $20

- Use: Electronic door locks, Cabinets, Access control

WATER SOLENOID VALVE

- Controls water flow ON/OFF
- Available in NC / NO type
- Easy installation
- Common sizes: 1/8", 1/4", 3/8", 1/2"

Voltage: 12V DC | Price: $10 – $20

- Use: Water dispensers, RO purifiers, Irrigation, Coffee machines

AIR SOLENOID VALVE (2/2 WAY)

- Controls air flow ON/OFF
- 2-Port 2-Way Normally Closed
- Brass body
- High durability

Voltage: 12V DC / 24V DC | Price: $10 – $18

- Use: Pneumatic systems, Air compressors, Automation

AIR SOLENOID VALVE (5/2 WAY)

- 5-Port 2-Position
- Controls double acting cylinders
- Fast switching
- High flow capacity

Voltage: 24V DC | Price: $18 – $35

- Use: Industrial automation, Pneumatic actuators

NORMALLY CLOSED (NC) VALVE

NC – Normally Closed

- Closed when de-energized
- Opens when energized
- Safe for fail-close applications
- Brass / SS body options

Voltage: 12V DC / 24V DC | Price: $10 – $25

- Use: Water control, Fuel control, Safety shut-off systems

NORMALLY OPEN (NO) VALVE

NO – Normally Open

- Open when de-energized
- Closes when energized
- Ideal for fail-open applications
- Brass / SS body options

Voltage: 12V DC / 24V DC | Price: $10 – $25

- Use: Air release, Ventilation, Drain systems

PROPORTIONAL SOLENOID VALVE

- Variable flow control
- High precision
- Analog / PWM control
- Used in precise automation

Voltage: 24V DC | Price: $30 – $60

- Use: Flow regulation, Pressure control, Robotic systems

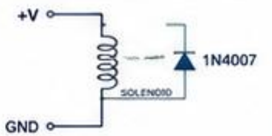
APPLICATION AREAS

Industrial Automation | Water Treatment & Irrigation | Security & Access Control | Automotive Systems | Medical Equipment | Home Appliances & Vending

TIPS FOR USE

- Always check voltage & current rating
- Use flyback diode for DC coils
- Choose NC or NO based on safety requirement
- Ensure proper mounting & orientation
- Use correct valve size for your application

DC SOLENOID WIRING (WITH FLYBACK DIODE)



Note: Prices are approximate (USD) and may vary by supplier, brand & quantity.

Essential Components for Your Projects

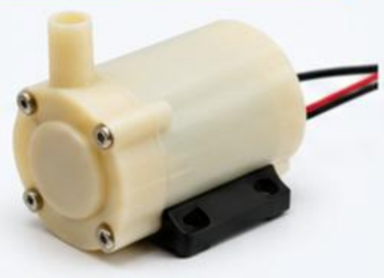
PUMPS & FANS

TYPES, FEATURES & APPLICATIONS

Works with Arduino, ESP32, Raspberry Pi

PUMPS

MINI WATER PUMP



Compact submersible pump ideal for small water transfer applications.

FEATURES

- Operating Voltage: 3V – 6V DC
- Flow Rate: 120 – 240 L/H
- Max Lift: 40 – 110 cm
- Current: 130 – 220 mA
- Size: 43 x 30 x 32 mm
- Quiet operation, lightweight

APPLICATIONS

- Small aquariums
- Water circulation
- DIY projects, science models

PRICE: $5 – $10

PERISTALTIC PUMP

Accurate low-volume fluid transfer without fluid contacting the pump mechanism.

FEATURES

- Operating Voltage: 6V – 24V DC
- Flow Rate: 0.1 – 300 ml/min
- Tubing Size: ID 1 – 4 mm
- Self-priming up to 2 m
- Reversible direction
- Chemical resistant tubing

APPLICATIONS

- Dosing chemicals
- Medical / lab equipment
- Water treatment, DIY robotics

PRICE: $12 – $25

DIAPHRAGM PUMP

High pressure pump suitable for continuous duty and various liquid transfer tasks.

FEATURES

- Operating Voltage: 12V / 24V DC
- Flow Rate: 1 – 5 LPM
- Max Pressure: 0.6 – 0.9 MPa
- Self-priming up to 2 – 3 m
- Dry run capable
- Durable and reliable

APPLICATIONS

- Spraying systems
- Car / bike wash
- Agriculture, misting, RO systems

PRICE: $15 – $30

FANS

5V BRUSHLESS FAN

Efficient low voltage fan for cooling electronic components and small enclosures.

FEATURES

- Operating Voltage: 5V DC
- Size: 50 x 50 x 10 mm
- Speed: 4000 – 6000 RPM
- Air Flow: 4 – 10 CFM
- Current: 0.12 – 0.20 A
- Low noise, energy efficient

APPLICATIONS

- Arduino / ESP32 projects
- Electronic cooling
- 3D printer, enclosures, DIY

PRICE: $3 – $8

12V COOLING FAN

High airflow fan for effective cooling in more demanding applications.

FEATURES

- Operating Voltage: 12V DC
- Size: 80 x 80 x 25 mm
- Speed: 2000 – 3000 RPM
- Air Flow: 25 – 40 CFM
- Current: 0.15 – 0.30 A
- Long life, strong airflow

APPLICATIONS

- PC / electronics cooling
- Power supplies
- Industrial / robotics projects

PRICE: $4 – $10

WHY USE PUMPS & FANS?

They help in fluid transfer, pressure generation, cooling and ventilation in countless DIY, industrial and automation applications.

GENERAL TIPS

- Use proper voltage & current supply
- Ensure tight connections
- For pumps: avoid dry running
- For fans: ensure proper airflow path

INTERFACING

Pumps & fans can be controlled using transistors, MOSFETs, relays or motor drivers with Arduino / ESP32.

PRICE NOTE

Prices are approximate (USD) and may vary based on brand, quality, region and supplier.

MAKE YOUR PROJECTS BETTER

Choose the right pump or fan for your needs and improve the performance and reliability of your projects!

HEATING / COOLING ACTUATORS

Convert Electrical Energy into Heat or Cooling for Embedded & IoT Applications

Used in: 3D Printers, Incubators, HVAC, Medical Devices, Lab Equipment, Smart Home, Automotive, Industrial Automation

Controlled by: Arduino, ESP32, Raspberry Pi, PLC etc.

1. HEATING ELEMENT

(Tubular Heater)

FEATURES

- High heating efficiency
- Fast heat generation
- Available in various power ratings (50W – 3kW+)
- Operates on AC or DC
- Robust and long life

TYPICAL APPLICATIONS

Water Heating, Ovens & Furnaces, 3D Printers, Industrial Machines

PRICE: $5 – $20 USD

2. CERAMIC HEATER

(PTC Ceramic Heater)

FEATURES

- PTC (Positive Temperature Coefficient)
- Self-regulating temperature
- Even heat distribution
- Compact & lightweight
- Safe, no open flame

TYPICAL APPLICATIONS

Incubators, Space Heaters, Dryers, Industrial Enclosures

PRICE: $8 – $20 USD

3. HEATER CARTRIDGE

(Cartridge Heater)

FEATURES

- High power density
- Fast response
- Available in various sizes (6mm – 20mm dia)
- Operates up to 600°C+
- Easy to install

TYPICAL APPLICATIONS

Hot Ends (3D Printers), Molds & Dies, Packaging Machines, Injection Molding

PRICE: $6 – $18 USD

4. PELTIER MODULE

(TEC1-12706)

FEATURES

- Solid state (no moving parts)
- Can be used for heating or cooling
- Compact and lightweight
- Operates on DC (12V typical)
- Temperature difference up to ~67°C

TYPICAL APPLICATIONS

Mini Coolers, Temperature Control, Electronics Cooling, Lab Equipment

PRICE: $5 – $10 USD

5. SILICONE HEATER PAD

(Silicone Heater Pad)

FEATURES

- Flexible and thin
- Uniform heat distribution
- Water & moisture resistant
- Easy to mount on flat or curved surfaces
- Operates on 12V / 24V

TYPICAL APPLICATIONS

3D Printer Heat Bed, Battery Warming, Drum / Tank Heating, Medical Devices

PRICE: $10 – $20 USD

6. THERMOELECTRIC COOLER (FAN + HEATSINK OPTIONAL)

(Peltier + Heatsink + Fan)

FEATURES

- Active cooling using Peltier effect
- Quick temperature reduction
- Requires heatsink & fan for better performance
- Compact & reliable
- Operates on DC

TYPICAL APPLICATIONS

Electronics Cooling, Mini Fridge / Cooler, Laser Diode Cooling, DIY AC / Chillers

PRICE: $10 – $25 USD

GENERAL NOTES

- ✔ Select actuator based on required temperature, power, size, and environment.
- ✔ Use appropriate drivers / relay / MOSFET / PWM modules for safe control.
- ✔ Always use thermal insulation and safety protection (fuse, thermostat, heat sink etc.).
- ✔ Ensure proper ventilation for cooling actuators.

CONTROL METHODS

- PWM Control (For Power Regulation)
- MOSFET / SSR (High Power Switching)
- Temperature Sensor + PID Control
- Fan / Heatsink (For Better Heat Dissipation)

QUICK COMPARISON

Actuator	Type	Operating Voltage	Max Temp / ΔT	Power Range	Best For
Heating Element	Resistive	AC 110V / 220V	Up to 800°C+	50W – 3000W+	High power heating
Ceramic Heater	PTC	AC / DC (12V – 240V)	Up to 300°C	20W – 200W	Safe, self-regulating heat
Heater Cartridge	Resistive	12V / 24V / 220V	Up to 600°C+	30W – 200W	Compact high temp heating
Peltier Module	Thermoelectric	DC 5V – 15V	ΔT up to ~67°C	30W – 60W	Heating or Cooling
Silicone Heater Pad	Resistive	12V / 24V / 110V / 220V	Up to 200°C	10W – 200W	Flexible surface heating
TEC + Fan + Heatsink	Thermoelectric	DC 12V	ΔT up to ~67°C	30W – 60W	Active cooling

SAFETY FIRST! High temperatures and high power can be dangerous. Always follow electrical safety guidelines and use appropriate protection.

Use Fuse | Use Thermal Cut-off | Proper Insulation | Keep Away from Water